

**INTERPRETABLE SPARSE MODELING OF LONGITUDINAL SIGNALS VIA
CRITICAL-RANGE RECTIFICATION AND ANYTIME RULE COMPRESSION**

by

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ABSTRACT

INTERPRETABLE SPARSE MODELING OF LONGITUDINAL SIGNALS VIA CRITICAL-RANGE RECTIFICATION AND ANYTIME RULE COMPRESSION

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High-dimensional longitudinal data arise in clinical monitoring, industrial control systems, and other sensor-driven domains where outcomes are often governed by threshold-and-lag behavior. Traditional longitudinal workflows frequently depend on expert guessing to nominate candidate variables, lag windows, and threshold hypotheses, followed by repeated hypothesis testing over a limited set of manually specified relationships. While such approaches can be useful in narrow settings, they are often less robust in high-dimensional regimes because important interactions may be missed, multicollinearity can destabilize inference, and the resulting process can be labor-intensive and difficult to scale. This dissertation develops an end-to-end framework for interpretable sparse longitudinal modeling that addresses these limitations through three stages: (i) critical-range rectification, which converts each lagged continuous measurement into a $\{-1, +1\}$ indicator denoting whether it lies within a data-driven range observed during events; (ii) sparse model fitting using an L1-regularized logistic baseline on the rectified design; and (iii) anytime rule compression ("logic polishing"), which collapses small-magnitude coefficients to produce a compact m-of-K rule model tuned to maximize Youden's J at a chosen operating point. Studies on synthetic data with known ground-truth rules and on real-world datasets, including the HAI industrial control benchmark, show that rectification improves feature-and-lag attribution, reduces false positives, and can decrease end-to-end training time. A tractable theoretical analysis uses the zero-

threshold arcsin relation under joint normality to show how sign binarization can contract pairwise correlations, improve conditioning of the Gram matrix, and increase the likelihood of satisfying the LASSO irrepresentable condition. The dissertation quantifies the contribution of each pipeline stage through ablations, stability studies, and expanded benchmark comparisons. The central claim is that a rectification-first, theory-informed, and compression-aware pipeline provides a more reliable and operationally usable alternative for threshold-driven longitudinal feature discovery than either raw sparse fitting or manually hypothesized expert-only workflows.

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I dedicate my dissertation to my wife Danielle.

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TABLE OF CONTENTS

	Page
LIST OF TABLES	x
LIST OF FIGURES	xi
Chapter	
1. INTRODUCTION	1
1.1 PROBLEM.....	3
1.2 CONTRIBUTIONS.....	3
1.3 RESEARCH QUESTIONS	4
1.4 EVALUATION PRIORITIES	5
1.5 THESIS ORGANIZATION.....	6
2. BACKGROUND	8
2.1 L1-REGULARIZED MODELS AND FEATURE SELECTION	8
2.2 SUPPORT RECOVERY AND THE IRREPRESENTABLE CONDITION	9
2.3 LONGITUDINAL LAG EXPANSION AND STRUCTURED SPARSITY	9
2.4 INTERPRETABILITY AS A DESIGN REQUIREMENT	10
2.5 COMPETING APPROACHES	10
3. RELATED WORK.....	12
3.1 SPARSE REGULARIZATION FOUNDATIONS	12
3.2 SUPPORT RECOVERY UNDER DEPENDENCE	13
3.3 GROUP, BLOCK, AND ORDERED SPARSITY	14
3.4 CORRELATION-AWARE AND ALTERNATIVE FEATURE SELECTION FAM- ILIES	15
3.5 INTERPRETABLE MODELING AND RULE-LEARNING LITERATURE	16
3.6 LONGITUDINAL HIGH-DIMENSIONAL MODELING CONTEXT	16
3.7 METHOD LINEAGE AND DISSERTATION POSITIONING	17
3.8 SUMMARY OF GAPS IN PRIOR WORK.....	18
4. APPROACH.....	20
4.1 WHY A REPRESENTATION-FIRST PIPELINE	20
4.2 METHOD INTUITION	21
4.3 PIPELINE STRUCTURE	22
4.4 HOW THE THEORETICAL CASE IS BUILT.....	24
4.5 HOW THE APPROACH MAPS TO THE RESEARCH QUESTIONS	26
5. RECTIFICATION IMPROVES SPARSE LONGITUDINAL SELECTION (RQ1)	27
5.1 A SIMPLE EXAMPLE TO AID INTUITION DEVELOPMENT.....	27
5.2 RQ1 EVALUATION FRAMEWORK.....	31

	Page
5.3 SYNTHETIC EVIDENCE WITH KNOWN GROUND TRUTH.....	34
5.4 COMPARISON TO COMPETING FEATURE-SELECTION BASELINES	37
5.5 REAL-WORLD EVIDENCE AND INTERPRETABILITY TRADEOFFS	40
5.6 ANSWER TO RQ1	49
6. THEORETICAL ANALYSIS (RQ2).....	53
6.1 WHY RQ2 MATTERS AFTER RQ1 EVIDENCE	53
6.2 IC IN COMPUTER-SCIENCE TERMS	54
6.3 TRACTABLE SETUP AND ASSUMPTIONS	54
6.4 WHY SIGN BINARIZATION CAN REDUCE DEPENDENCE	55
6.5 LEMMA PATHWAY (WITH INTUITION)	56
6.6 MAIN THEOREM (PROBABILITY FORM)	57
6.7 PROOF SKETCH	58
6.8 PRACTICAL INTERPRETATION	59
6.9 A SMALL INTERFERENCE EXAMPLE	59
6.10 HOW THIS DIFFERS FROM PENALTY-ONLY FIXES.....	60
6.11 WHAT THIS THEOREM DOES NOT CLAIM.....	61
6.12 EMPIRICAL BOUNDARY CHECKS BEYOND THE THEOREM	61
6.13 EMPIRICAL IMPLICATIONS OF RQ2.....	62
6.14 PRACTICAL SIGNIFICANCE	62
7. ANYTIME RULE COMPRESSION (RQ3).....	64
7.1 WHY RQ3 MATTERS AFTER RQ1 AND RQ2	64
7.2 FROM RECTIFIED SPARSE MODELS TO RULE SPACE.....	65
7.3 ANYTIME COMPRESSION OBJECTIVE.....	65
7.4 ALGORITHMIC PIPELINE	66
7.5 WHY THE METHOD IS “ANYTIME”	67
7.6 EMPIRICAL RESULTS.....	67
7.7 STATISTICAL FRAMING OF “NO MATERIAL DEGRADATION”	68
7.8 RELATIONSHIP TO ADJACENT INTERPRETABLE METHODS	69
7.9 MOTIVATION AND APPLICATION CONTEXT	69
7.10 LIMITATIONS AND SCOPE.....	70
7.11 ANSWER TO RQ3	70
8. FUTURE WORK.....	72
8.1 EXTENDING THE THEORETICAL SCOPE.....	72
8.2 BROADER EMPIRICAL VALIDATION	72
8.3 RICHER RULE FAMILIES AND HYBRID MODELS.....	73
8.4 PROSPECTIVE DEPLOYMENT AND HUMAN EVALUATION	73
8.5 TOOLING AND REPRODUCIBILITY.....	73
8.6 CLOSING PERSPECTIVE.....	73
9. CONCLUSION	75

	Page
9.1 RESEARCH PROGRAM SUMMARY	75
9.2 CONCLUSIONS BY RESEARCH QUESTION.....	76
9.3 PRIMARY DISSERTATION CONTRIBUTIONS.....	77
9.4 POSITIONING RELATIVE TO PRIOR WORK	78
9.5 SCOPE BOUNDARIES AND LIMITATIONS	78
9.6 PRACTICAL IMPLICATIONS	79
9.7 FINAL STATEMENT	79
REFERENCES	80
APPENDICES	
A. SYNTHETIC DATA DEVELOPMENT	86
B. LEMMAS	90
B.1 LEMMA 1	90
B.2 LEMMA 2	92
B.3 LEMMA 3	95
B.4 LEMMA 4	98
VITA	104

LIST OF TABLES

Table	Page
1. Synthetic data case study results.	35
2. RQ1 repeated-resample stability and ablation summary.	37
3. RQ1 expanded synthetic baseline benchmark.	43
4. Real-world transformed vs. untransformed results.	44

LIST OF FIGURES

Figure	Page
1. Threshold-and-Lag.	7
2. Toy Example Coefficient Vectors.	28
3. Intuition Developement for One Relevant Curve.	32
4. Intuition Developement for Two Relevant Curves.	33
5. Synthetic Data Transformed vs. Untransformed.	36
6. RQ1 ablation summary.	38
7. RQ1 pairwise support stability.	39
8. RQ1 selection-frequency heatmaps.	40
9. RQ1 runtime scaling under lag expansion.	41
10. Katrutsa and Strijov Method Results.	42
11. HAI Results.	45
12. HAI ROC.	46
13. HAI J vs. K.	47
14. RQ1 cross-domain transfer deltas.	48
15. Ionosphere protocol distinction.	49
16. Goose Bay repeated-split protocol audit.	50
17. Goose Bay quantile sensitivity.	51
18. Goose Bay penalty sensitivity.	52
19. Anytime Compression (RQ3) Results.	68
20. Synthetic Data Generation Step 1.	87
21. Synthetic Data Generation Step 2.	87
22. Synthetic Data Generation Step 4.	88

23. Synthetic Data Generation Step 7, Curve Flattening.....	89
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CHAPTER 1

INTRODUCTION

Longitudinal data contain repeated measurements of the same variables across time and are central to many scientific and operational problems. Examples include symptom progression in clinical cohorts, temporal variation in biological markers, and multi-wave surveys of human decision behavior [8, 29, 37]. Across these settings, outcomes are often triggered by threshold-and-lag mechanisms: a variable enters a critical range, then a target event appears after a delay as with Figure 1. This structure is common in sensing and monitoring contexts where decision-makers need not only accurate predictions, but also clear explanations of which variables and lags drove the prediction.

Specifically, several study areas that have shown a tendency to involve threshold-and-lag scenarios are:

- **Ecosystem Dynamics.** Ecosystems often exhibit threshold-dependent responses with lagged effects, such as delayed population shifts [7].
- **Epidemic Spreading.** The spread of infectious diseases depends on multiple thresholds (e.g., transmission, recovery, and vaccination rates) [15].
- **Control Systems.** Many engineering systems feature threshold-driven lagged behaviors in response to control inputs or environmental changes [22].
- **Financial Markets.** Market reactions to economic indicators, interest rates, or sentiment often follow threshold effects with delayed responses [32].

In high-dimensional longitudinal modeling, a standard strategy is to construct a lag-expanded design matrix and apply a sparse estimator. This strategy is attractive because sparse models can retain predictive performance while reducing dimensionality [18, 33]. However, lag expansion also introduces strong dependence among adjacent lags and among correlated channels, creating exactly the regimes where support recovery becomes unstable and model-selection assumptions can fail [5, 41]. Practical outcomes include arbitrary swapping among correlated predictors, sensitivity to tuning choices, and weak reproducibility of selected supports across resamples.

A substantial body of work addresses these issues through penalty design and optimization advances, including elastic net, adaptive lasso, group-based penalties, and ordered lag constraints [16, 34, 40, 43, 44]. These methods provide important baselines and often improve behavior relative to plain lasso, but they still operate directly on raw correlated representations and may not provide explicit threshold semantics for human review. Correlation-aware variants in domain-specific contexts further reinforce this point: dependence can be mitigated, but stable, interpretable attribution remains difficult when predictors are densely coupled [11, 38, 42].

A second challenge is interpretability under high-stakes use. Post hoc explanations of complex models can be useful, but interpretable-model-first approaches are often preferable when transparency, auditability, and operational trust are mandatory [17, 25]. Rule-structured models are especially relevant because they align with human review workflows and can be validated directly by domain experts [1–3]. This dissertation therefore focuses on an inherently interpretable sparse pipeline rather than black-box prediction with after-the-fact explanation.

This approach builds from prior work in this research line: a rectification-first longitudinal feature-selection method, a scoped theoretical analysis linking binarized transformation to improved irrepresentable-condition (IC) behavior, and an anytime rule-compression method that con-

verts sparse models into compact logic-like forms [19–21]. The dissertation unifies these components into one framework, strengthens the supporting analysis, broadens empirical evaluation, and emphasizes reproducible implementation.

1.1 PROBLEM

This work addresses three tensions that arise jointly in longitudinal feature learning:

- Correlation + high dimension $\rightarrow$ unstable sparse support selection and weak lag attribution.
- Operational use $\rightarrow$ limited compute budgets and constraints on model complexity.
- High-stakes deployment $\rightarrow$ demand for transparent, auditable rules instead of opaque scoring functions.

The central research question is whether representation-level rectification can make sparse longitudinal selection more reliable before model fitting, and whether a principled post-selection compression step can preserve discrimination while improving human interpretability. In contrast to approaches that rely only on penalty modifications, this dissertation investigates a preprocessing-plus-selection-plus-compression pipeline designed to explicitly encode threshold behavior, reduce correlation burden, and expose causal hypotheses at the feature-lag level.

Thesis statement: critical-range rectification improves reliability of sparse longitudinal feature recovery by contracting harmful dependence structures in lag-expanded data, and anytime rule compression converts resulting sparse models into compact, operationally usable rule forms with minimal loss in discriminative performance [19, 20]. The scope emphasizes inherently interpretable sparse models with explicit attribution; deep learning appears only as contextual baselines where needed.

1.2 CONTRIBUTIONS

This dissertation makes four integrated contributions.

1. **A unified rectification-first longitudinal pipeline.** Building on prior conference work, the dissertation consolidates critical-range transformation, sparse logistic selection, and downstream rule extraction into a single end-to-end framework [20, 21].
2. **Theory-informed analysis of correlation and selection behavior.** The dissertation extends tractable regime arguments relating transformed representations to improved support-recovery conditions, situating these results against established lasso consistency theory and dependence-aware critiques [5, 20, 41].
3. **Anytime compression to interpretable m-of-K style rules.** Sparse coefficient models are converted into compact rule forms that can be audited and deployed under time and complexity constraints, extending recent rule-compression results [1, 19].
4. **A broad empirical protocol for longitudinal evaluation.** The dissertation compares raw and rectified representations across synthetic and real longitudinal benchmarks, including signal-rich settings such as ICS anomaly data and ionospheric radar returns, with evaluation focused on both discrimination and attribution quality [27, 28].

1.3 RESEARCH QUESTIONS

This dissertation is organized around three research questions that move from empirical utility to theoretical credibility and then to deployment usability:

- **RQ1:** Can critical-range rectification produce a stable sparse baseline with reliable feature and lag attribution?
- **RQ2:** Why does rectification work when it works, and how far can that mechanism be generalized beyond the current theoretical assumptions?
- **RQ3:** Can the resulting sparse solution be compressed into compact logical rules without unacceptable loss in operational discrimination?

Together, these questions are intentionally sequenced. There is little value in compressing a model into rules if upstream sparse attribution is unstable, and little scientific value in empirical gains if the mechanism producing those gains is not understood. Accordingly, RQ1 tests whether representation-level rectification improves sparse longitudinal selection in practice; RQ2 explains why those improvements should occur and where the theoretical boundary lies; and RQ3 asks whether the recovered sparse structure can be converted into models that are usable in real decision workflows [5, 17, 25, 41].

1.4 EVALUATION PRIORITIES

Across all three research questions, the dissertation evaluates the proposed framework using a common set of priorities:

- **Predictive discrimination:** metrics such as AUC and Youden’s J at operational thresholds, interpreted in dataset context [20, 27, 28].
- **Attribution quality and stability:** support recovery behavior, lag localization fidelity, and stability across folds or resamples.

- **Interpretability complexity:** active-feature count, rule count, and rule length as direct proxies for human audit burden [1, 17].
- **Efficiency:** end-to-end training and inference time, including preprocessing and compression overhead.
- **Practical non-inferiority where needed:** when compressed rules are compared against upstream sparse baselines, the evaluation uses interval-based equivalence logic to determine whether performance loss is practically small [26].

1.5 THESIS ORGANIZATION

Following this introduction, the dissertation is organized as follows. The background chapter formalizes longitudinal feature-learning assumptions, sparse regularization foundations, and interpretability criteria, and the related-work chapter positions competing methods in grouped, ordered, dependence-aware, and rule-based selection [6, 31, 39]. The approach chapter then presents the rectification-first pipeline in unified notation, explains the intuition behind its design, and describes the theoretical pathway used later to justify the central claims.

The subsequent chapters address the three research questions directly: empirical evidence for rectification, theoretical analysis of the mechanism, and anytime rule compression for deployment-ready interpretability. The final chapters summarize limitations, future extensions, and deployment implications for longitudinal decision support. Proof details are discussed at a high level in the main text and provided step-by-step in the appendices.

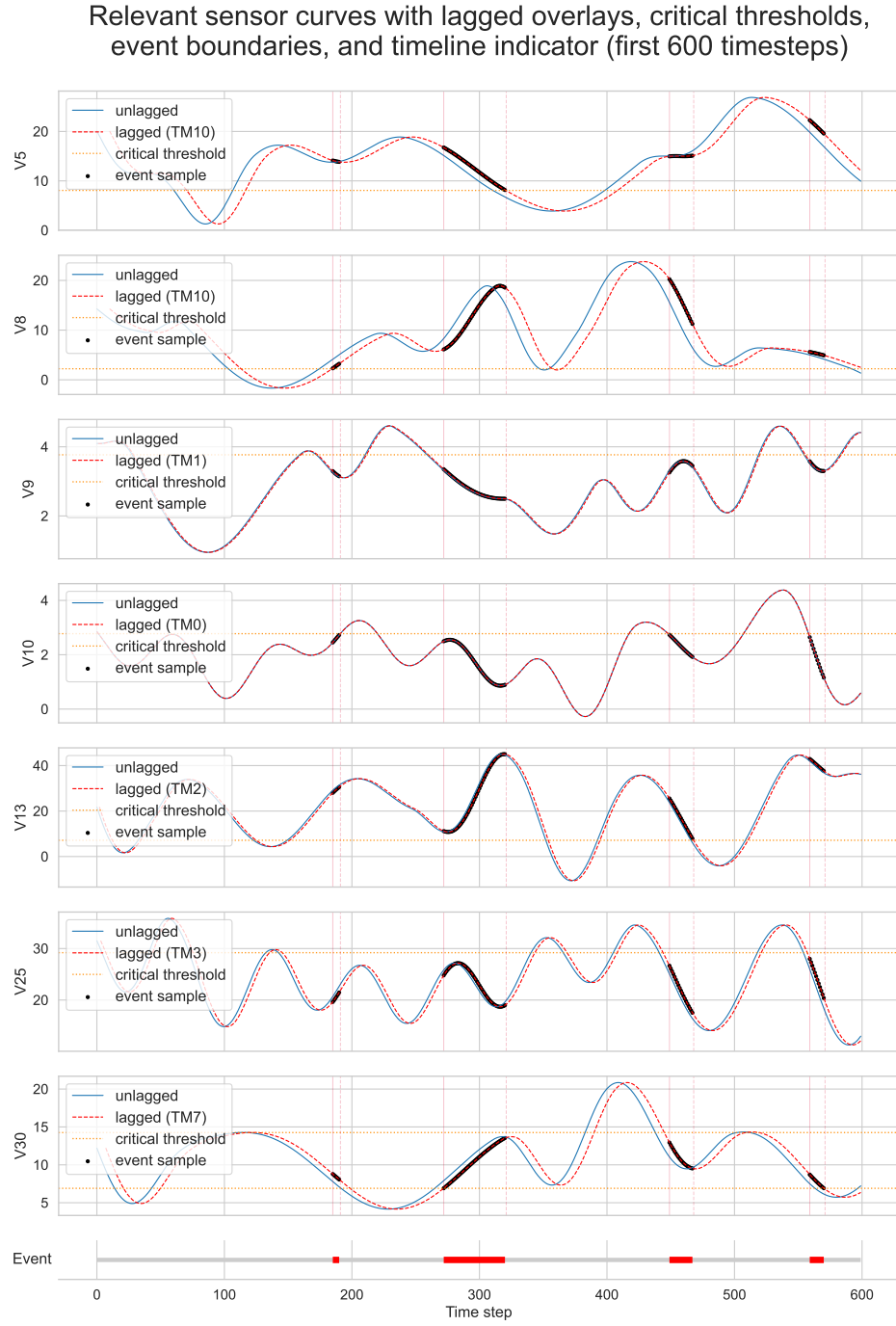


Figure 1. Threshold-and-lag behavior in longitudinal signals using the actual synthetic data generated for the experiments discussed later in the document.

CHAPTER 2

BACKGROUND

Sparse modeling with L1 penalties is a standard approach for feature selection in high-dimensional data because it often yields compact models with practical predictive performance [18, 33]. In longitudinal settings, sparse selection is also used for lag attribution: selected coefficients identify not only which variables matter, but when they matter. This chapter reviews the background needed for the dissertation, including sparse regularization, support-recovery theory under dependence, interpretability requirements, and competing method families.

Longitudinal applications motivate this framing because repeated-measure data often contain delayed and heterogeneous effects across people, systems, or environments [8, 29, 37]. When temporal predictors are expanded across lags, model dimensionality and dependence both increase, so variable selection and interpretability must be considered jointly rather than as separate post-processing steps.

2.1 L1-REGULARIZED MODELS AND FEATURE SELECTION

The LASSO and L1-regularized logistic regression induce sparsity by shrinking many coefficients to exactly zero, creating an embedded feature-selection mechanism inside model fitting [33]. This property is one reason L1 methods are widely used in longitudinal pipelines: selected nonzero lag terms can provide concise, testable hypotheses about delayed effects.

Modern implementations make these methods computationally practical at scale: coordinate-descent path algorithms with warm starts support efficient generalized linear modeling across many

tuning values [6, 31]. Beyond plain L1 penalties, elastic net, adaptive lasso, and group-aware variants address correlated and block-structured predictors, which is directly relevant for lagged longitudinal features [12, 16, 40, 43, 44].

2.2 SUPPORT RECOVERY AND THE IRREPRESENTABLE CONDITION

For feature selection, prediction accuracy alone is insufficient; the key question is whether selected supports match the true active set. Zhao and Yu formalized this issue through the irrerepresentable condition (IC), showing that support recovery consistency for lasso depends critically on covariance structure [41]. When off-support predictors are too correlated with active predictors, sparse recovery can fail even when predictive fit appears strong.

This limitation is central in lag-expanded longitudinal designs, where adjacent lags and related channels are frequently collinear. Reviews of lasso under dependence consistently report this selection-versus-prediction tension, with false discoveries and unstable supports common under strong correlation [5]. Correlation-aware variants, including transformed/weighted and rank-based approaches, can improve robustness in specific regimes, but they do not eliminate the underlying dependence challenge in general [23, 42].

From a linear-algebra perspective, the issue is closely tied to conditioning and inverse-Gram stability, which govern how sensitive selected supports are to perturbations in data and tuning [9]. For this reason, the dissertation treats representation-level correlation contraction as a first-class objective before sparse fitting, and uses IC-oriented reasoning as the theoretical lens for interpreting support behavior.

2.3 LONGITUDINAL LAG EXPANSION AND STRUCTURED SPARSITY

Lag expansion is the dominant way to represent temporal effects in classical supervised learning, but it can quickly create high-dimensional, redundant design matrices. Ordered and structured penalties attempt to encode temporal assumptions directly in the coefficient space, for example monotone lag decay or shared supports across tasks [34, 35].

Additional group-penalized optimization work improves runtime and practical convergence for non-orthonormal designs, making structured sparse models more usable in real pipelines [39]. These methods are important comparators for this work: they preserve the original numeric representation and optimize increasingly sophisticated penalties, whereas the proposed pipeline changes representation first and then applies sparse selection.

2.4 INTERPRETABILITY AS A DESIGN REQUIREMENT

In high-stakes settings, interpretability is not an afterthought: end-users must understand why a prediction was made and be able to verify it against domain knowledge. Interpretable-model-first arguments emphasize that transparent model classes are often preferable to post hoc explanations layered on top of black boxes [17, 25]. This principle is especially relevant when longitudinal decisions affect operations, safety, or clinical workflows.

Rule-based models provide a natural interface for such settings because they are directly inspectable and can be validated by experts testing each feature and lag individually against specific rules. Classical logical-analysis frameworks and modern optimal-rule-list methods show that transparent rule structures can be competitive while preserving auditability [1–3]. This work adopts this perspective by treating rule compression as part of model construction, not merely post-hoc explanation.

2.5 COMPETING APPROACHES

Related works are analyzed contextually with this work in greater detail in the next chapter, but a brief overview of the relevant competing families include:

- Group and structured sparsity methods impose constraints on coefficients, such as group-wise inclusion or ordered lag decay; they often improve stability but generally do not alter the raw input dependence structure [16, 34, 39, 40].
- Dependence-aware sparse variants attempt to improve selection under strong correlation through reweighting, transformed objectives, or robust/rank formulations [11, 23, 42].
- Wrapper and filter selection methods remain pragmatic baselines, especially in domain pipelines where model families vary, but they are often heuristic and may provide weaker attribution consistency across resamples [13, 14].
- Robust multicollinearity-focused estimators and heuristics can improve fit under contamination and ill-conditioning, but may prioritize prediction stability over sparse interpretability [24].
- Logic-rule and LAD-style methods explicitly target threshold semantics and human-readable decisions, yet can face combinatorial scaling pressures as feature spaces grow [1–3].

Taken together, this literature motivates the design choice for this work: combine representation-level preprocessing with sparse estimation and rule-oriented compression. This places the work between two extremes, pure penalty engineering on raw lag-expanded data and direct combinatorial rule search, with the goal of balancing reliability, transparency, and computational feasibility.

CHAPTER 3

RELATED WORK

This chapter reviews method families most relevant to the problem identified in this work: sparse longitudinal feature selection under strong dependence, with interpretable downstream decision logic. The discussion is organized to move from core sparse-learning foundations to structured penalties, dependence-aware alternatives, and rule-based interpretability frameworks. The final sections position this work relative to prior work in the same method lineage.

3.1 SPARSE REGULARIZATION FOUNDATIONS

L1-regularized modeling is the canonical starting point for sparse feature selection. The lasso formulation introduced by Tibshirani established a convex mechanism that performs shrinkage and variable selection simultaneously, making it practical for high-dimensional settings [33]. In classification settings, L1-regularized logistic regression inherits the same sparsity behavior and is widely used when model compactness and feature attribution are important.

Early theoretical and empirical analyses also clarified why L1 penalties are often preferable to purely L2-regularized approaches when many predictors are irrelevant. In sparse-relevance regimes, L1 methods can scale better with growing nuisance dimensionality than rotationally invariant alternatives [18]. This intuition remains central for longitudinal lag-expanded designs, where many candidate lag terms may not be causally relevant.

A second reason this family became dominant is computational maturity. Coordinate-descent path algorithms provide efficient fitting across full regularization paths for generalized linear mod-

els and related objectives [6, 31]. These advances made sparse modeling feasible at operational scales and enabled cross-validated tuning in realistic workflows rather than one-off, expensive optimization.

However, plain lasso is not the endpoint of sparse modeling. Elastic net and adaptive lasso were proposed to improve behavior under correlation and to reduce some forms of selection bias [43, 44]. This work treats these methods as important baselines, but not as complete solutions for threshold-and-lag attribution in heavily collinear longitudinal representations.

3.2 SUPPORT RECOVERY UNDER DEPENDENCE

Predictive accuracy and correct feature recovery are distinct goals. Zhao and Yu formalized this distinction by showing that lasso model-selection consistency depends on covariance-structure conditions, particularly the irrepresentable condition (IC) [41]. In practical terms, when inactive predictors are too correlated with active predictors, support recovery may fail even if prediction remains competitive.

This issue is acute in lag-expanded longitudinal data, where neighboring lags and related channels often have strong dependence. Recent reviews emphasize that lasso and many derivatives can produce unstable supports and elevated false positives in dependent designs, especially under common tuning choices optimized for prediction rather than exact support recovery [5].

Dependence-aware variants address parts of this problem. Examples include transformed or weighted formulations for highly correlated predictors and robust rank-based alternatives that reduce sensitivity to link-function misspecification or heavy-tailed behavior [23, 42]. These methods improve robustness in specific regimes, but they generally do not provide a unified threshold semantics and lag-rule representation for operational interpretation.

From a matrix-analysis viewpoint, these selection instabilities reflect conditioning and inverse-Gram sensitivity, which directly affect sparse-support perturbation behavior [9]. This perspective motivates representation-level interventions before sparse fitting, rather than relying only on penalty redesign in the original feature space.

3.3 GROUP, BLOCK, AND ORDERED SPARSITY

Structured penalties were developed to stabilize selection when predictors have known organization. Early simultaneous-selection formulations for multiresponse models and later grouped penalties introduced mechanisms that select predictors at the block level rather than coefficient-by-coefficient [35, 40]. This is relevant in longitudinal settings where features can be grouped by channel, basis expansion, or lag family.

Subsequent work extended grouped sparsity to logistic settings and improved practical optimization. Group lasso for logistic regression, blockwise sparse regression, and unified majorization-descent solvers significantly improved feasibility for high-dimensional non-orthonormal designs [12, 16, 39]. These methods remain strong comparators because they encode structural assumptions directly in the objective while retaining convexity in many cases.

Temporal structure has also been added explicitly through ordered penalties. Ordered lasso introduces monotonicity constraints across lag coefficients, reflecting assumptions such as decaying lag influence and yielding interpretable lag profiles when those assumptions hold [34]. This is a particularly relevant baseline for relevant experiments because it targets time-lagged sparsity directly.

Despite these advances, structured penalties still operate in the original correlated representation. They can improve stability and interpretability relative to plain lasso, but they do not

inherently convert continuous trajectories into explicit threshold logic. This work positions its rectification step as complementary to these methods: change the representation first, then apply sparse learning.

3.4 CORRELATION-AWARE AND ALTERNATIVE FEATURE SELECTION

FAMILIES

Beyond penalized linear models, several approaches attempt to reduce redundancy through direct relevance-redundancy criteria. Katrutsa and Strijov proposed a quadratic-programming feature-selection framework where pairwise similarity and target relevance are balanced in a convex relaxation [11]. This strategy can reduce redundant supports, but it is primarily pairwise in construction and does not natively express lag-threshold rules.

Robust multicollinearity-focused estimators provide another pathway. Methods combining robust regression ideas with conditioning-oriented penalties can improve estimation under contamination and ill-conditioning [24]. These approaches are useful for robustness analysis but typically prioritize fit stability over compact, human-auditable sparse logic.

Filter, wrapper, and embedded selection families remain widely used in practice because they are flexible and model-agnostic [14]. Recent hybrid ranking-plus-search approaches also show good empirical performance in domain-specific pipelines [13]. Still, these families can be sensitive to search heuristics, may yield unstable subsets under resampling, and usually do not produce direct temporal-threshold semantics without additional modeling layers.

Accordingly, this work includes these methods primarily as conceptual comparators rather than as primary interpretability baselines. Their strengths are pragmatic feature screening and broad applicability; their limitations are weaker guarantees for sparse lag attribution and logic-level in-

interpretability.

3.5 INTERPRETABLE MODELING AND RULE-LEARNING LITERATURE

Interpretability-focused literature increasingly argues that high-stakes deployments should prefer inherently interpretable models over post hoc explanations of black boxes whenever performance is competitive [17, 25]. This framing strongly influences the design choices in this work: interpretability is treated as a primary objective, not a secondary (post-hoc) reporting artifact.

Rule-based models are central to this viewpoint. Classical Logical Analysis of Data (LAD) frameworks formalize binarization and logic-pattern extraction, including optimization-based treatment of cut-point selection and pattern construction [2, 3]. These methods establish the value of explicit threshold logic for decision support, but can face combinatorial scaling limits in large, highly correlated feature spaces.

Modern rule-list methods demonstrate that transparent models can be both accurate and principled when optimization is carefully designed. Certifiably optimal rule-list learning shows that exact or bounded-search approaches can produce compact, auditable models with competitive predictive performance on structured tasks [1]. This work does not replicate global rule-list optimization, but it draws from the same interpretability objective and complexity-aware evaluation philosophy.

Relative to this literature, the strategy for this work is to anchor interpretability in sparse model fitting and then compress coefficients into logic-like forms through an "anytime" procedure. This avoids full combinatorial search while still yielding compact rule structures suitable for expert review.

3.6 LONGITUDINAL HIGH-DIMENSIONAL MODELING CONTEXT

Longitudinal methods in broader statistics and machine learning emphasize that repeated measures introduce heterogeneity, temporal dependence, and subgroup structure that can invalidate cross-sectional assumptions. Model-based clustering with regularized mixed-effects formulations demonstrates one advanced path for high-dimensional trajectory modeling [38]. These methods are powerful for discovering latent trajectory classes, but they optimize a different end goal than sparse, explicit lag-rule extraction.

Domain literature further illustrates why interpretable longitudinal modeling matters. Clinical and biological longitudinal studies report heterogeneous temporal pathways and delayed outcome behavior, reinforcing the need for methods that preserve temporal attribution [8, 37]. Similar heterogeneity appears in socio-environmental longitudinal surveys [29]. These contexts motivate models that can be inspected and discussed by domain experts rather than only ranked by black-box metrics.

Benchmark infrastructure also matters for evaluating longitudinal methods. Public anomaly and signal datasets used in this line of inquiry, such as HAI ICS telemetry and ionospheric radar-return data, provide realistic stress tests for lag attribution, sparse recovery, and interpretability tradeoffs [27, 28]. Related engineering contexts with threshold-driven transitions likewise support the relevance of range-based temporal reasoning [10].

3.7 METHOD LINEAGE AND DISSERTATION POSITIONING

This work is not a stand-alone methodology detached from prior work; it extends a concrete sequence of methods developed in earlier publications. The initial conference work introduced a rectification-first perspective for longitudinal feature learning, where critical-range binarization precedes sparse selection [21]. This established the basic preprocessing-plus-lasso pipeline and

empirical motivation.

Subsequent work added a scoped theoretical analysis and broader case studies. The 2025 efficient longitudinal feature-selection paper linked binarized transformation to correlation contraction arguments and improved IC-related behavior under stated assumptions, while also reporting computationally practical implementations [20]. This paper provides the main theoretical bridge between representation design and support recovery.

The companion 2025 anytime rule-compression paper addressed the remaining interpretability gap: sparse coefficients are often still too numerous for direct human use. The anytime compression stage converts sparse rectified models into compact m-of-K style rule behavior with controlled discrimination tradeoffs and early-stop practicality [19]. This contribution aligns the pipeline with interpretable-model deployment requirements discussed earlier in this chapter.

Compared with related work, the dissertation’s distinguishing emphasis is integration. Rather than proposing only a new penalty, only a new binarizer, or only a new rule learner, it unifies all three stages and evaluates them as one longitudinal decision-support pipeline. The key comparative hypothesis is that representation-level rectification plus sparse selection plus anytime compression better balances support reliability, interpretability, and compute constraints than any single-stage alternative.

3.8 SUMMARY OF GAPS IN PRIOR WORK

The reviewed literature reveals four persistent gaps that motivate this dissertation.

- Sparse penalties are computationally mature and often predictive, but support consistency remains fragile under strong lag-induced dependence [5, 41].

- Structured and dependence-aware penalties improve behavior in specific regimes, yet usually remain tied to raw correlated representations and do not produce threshold-native logic outputs [34, 40, 42].
- Rule-learning frameworks deliver transparency, but direct combinatorial optimization can be difficult to scale in high-dimensional longitudinal spaces [1, 2].
- Prior work in this dissertation line introduced the key pieces, but an end-to-end, reproducible, and extensively benchmarked synthesis is still needed [19–21].

The remainder of the dissertation addresses these gaps through unified methodology, theory-informed analysis, and comparative empirical evaluation.

CHAPTER 4

APPROACH

This chapter describes the research approach used throughout the dissertation. The central design choice is to intervene at the representation level before sparse optimization and then to compress the resulting sparse model into an explicit rule form. The overall pipeline is therefore: (i) lag-expand longitudinal measurements, (ii) rectify each candidate feature into a critical-range indicator, (iii) fit a sparse logistic baseline on the rectified representation, and (iv) compress the active coefficients into a compact m -of- K style rule when operational interpretability is required [19–21].

The chapter serves two purposes. First, it explains the intuition behind the method family as an integrated system rather than as three disconnected tricks. Second, it shows how the theoretical case is constructed: begin with an observable empirical failure mode in lag-expanded sparse selection, propose a representation-level intervention aimed at that failure mode, and then formalize the mechanism through a tractable argument tied to the irrepresentable condition (IC).

4.1 WHY A REPRESENTATION-FIRST PIPELINE

The motivating problem is not simply that longitudinal data are high dimensional. It is that lag expansion creates many predictors that are near-duplicates, shifted copies, or correlated surrogates of one another. In that regime, sparse penalties can still yield good prediction, but support recovery and lag attribution often become unstable [5, 41]. A method that only optimizes predictive fit can therefore produce models that are difficult to trust scientifically or operationally.

The dissertation’s approach is to alter the feature space before fitting. Rather than asking the sparse learner to disentangle raw continuous trajectories directly, the method first asks a simpler question of each feature-lag candidate: was the measurement inside or outside a range that has historically been associated with events? This turns continuous temporal signals into logical indicators that are closer to the threshold-and-lag semantics of the target problem. Sparse fitting then operates on a representation in which irrelevant amplitude variation is suppressed and the feature-selection task is more directly aligned with event logic.

This decision also reflects a practical systems argument. Representation redesign and penalty redesign solve different problems. Penalty engineering changes optimization geometry in coefficient space. Rectification changes statistical geometry in feature space before optimization. The dissertation emphasizes the second option because it can be layered onto mature sparse-learning software without requiring a custom solver, while still targeting the structural dependence issue that makes raw sparse supports unreliable.

4.2 METHOD INTUITION

4.2.1 Threshold-And-Lag Intuition

The intended problem class is one in which events are triggered when one or more variables enter critical operating ranges and the observed outcome may follow after a delay. In those settings, the exact numeric amplitude of a measurement is often less important than whether it crossed into the event-associated region at the relevant lag. Rectification encodes that event logic directly.

This is the core intuition behind the dissertation’s methods: if the phenomenon of interest is effectively governed by threshold conditions, then a representation that marks threshold satisfaction

should make the true causal structure easier to recover than a representation that preserves every continuous fluctuation. The rectified representation does not claim to preserve all information. It intentionally trades some magnitude detail for clearer logical contrast.

4.2.2 Toy-Example Intuition

The toy examples used later in the dissertation illustrate this idea in a controlled form. When raw continuous features are moderately correlated, lasso can distribute weight across irrelevant but similar variables. After binarization, the relevant features more closely match the event pattern while many irrelevant features collapse into less informative indicator sequences. In effect, the search problem shifts from “which continuous columns best interpolate the outcome?” to “which threshold indicators best explain the event pattern?” That shift is what motivates the full pipeline.

4.2.3 From Sparse Coefficients to Usable Decisions

Even when rectification improves support concentration, the resulting sparse model may still contain too many active coefficients for direct human use. A linear score with dozens of nonzero lag terms is better than a black box, but it is not yet an operational rule. The final stage of the approach therefore treats sparse fitting as an intermediate representation and compresses it into an explicit decision rule. This preserves the front-end efficiency of convex sparse learning while recovering a model form closer to how experts reason about triggers, audits, and deployment criteria.

4.3 PIPELINE STRUCTURE

4.3.1 Stage 1: Longitudinal Expansion and Critical-Range Rectification

Starting from longitudinal measurements, the first step is to create a lag-expanded design matrix so that each predictor corresponds to a specific variable-lag pair. For each candidate feature, the method then estimates a *critical range*: a range of observed values associated with event occurrence. The raw value is converted into a binary indicator that records whether the measurement falls inside or outside that range.

In the simplest analyzed setting this is equivalent to zero-threshold sign binarization after standardization, which makes the theory analytically tractable. In the practical pipeline used for experiments, the thresholds need not be zero and the active condition can correspond to one-sided or interval-like ranges learned from the data. The practical objective is the same in both cases: replace noisy continuous variation with a signal that directly encodes threshold satisfaction.

4.3.2 Stage 2: Sparse Baseline Fitting

Once the rectified design matrix is built, the dissertation fits an L1-regularized logistic regression model as the baseline sparse learner [6, 31, 33]. This stage answers which feature-lag indicators remain necessary once the representation has already encoded threshold logic. Nonzero coefficients then become candidate causal or operational triggers in the sense relevant to the target event.

The baseline is intentionally simple. More elaborate penalties exist and are treated as important comparators, but the dissertation’s claim is that a standard sparse learner becomes more reliable when the representation itself has been improved. This keeps the method modular and makes comparisons against other sparse baselines easier to interpret.

4.3.3 Stage 3: Anytime Rule Compression

The active sparse coefficients are then ordered by magnitude and compressed into progressively simpler rule candidates. Logic polishing assigns a common magnitude to retained terms, removes small residual coefficients, and evaluates nested prefixes as m -of- K decision rules [19]. Because each prefix yields a usable rule, the process is “anytime”: it can stop as soon as complexity is low enough and performance is still acceptable for the deployment context.

This stage is not an unrelated post-processing convenience. It is part of the dissertation’s overall research approach because the project is motivated by interpretable deployment, not only by feature-selection accuracy. The sparse model creates a focused candidate set; the compression stage converts that candidate set into a form that humans can audit, challenge, and implement.

4.4 HOW THE THEORETICAL CASE IS BUILT

4.4.1 Target of the Proof

The theoretical question is not whether every possible thresholding rule improves every sparse model. The goal is narrower and more defensible: show, in a tractable regime, that binarization can make sparse support recovery more likely by reducing harmful dependence patterns. The IC provides the right lens because it directly formalizes when inactive predictors can masquerade as active ones under lasso-style selection [41].

4.4.2 Proof Strategy

The proof structure used in this dissertation follows a simple progression.

1. Start with the observed empirical problem: lag-expanded predictors are strongly dependent, so sparse support recovery is unstable even when prediction is acceptable.

2. Introduce a tractable transformation: zero-threshold sign binarization under joint normal assumptions.
3. Use the arcsin relationship to show that pairwise correlation is contracted in the transformed representation.
4. Show that this contraction reduces inactive-active leakage and can improve conditioning of the active Gram block in the positive-correlation analyzed regime.
5. Translate those matrix effects into a higher probability of satisfying the IC, which in turn supports more reliable sparse support recovery.

This progression matters because it mirrors how the dissertation argues its case overall. The theory does not begin from an abstract theorem in search of an application. It begins from a concrete modeling failure mode, proposes a representation-level intervention for that failure mode, and then builds a mathematical structure that explains why the intervention should help.

4.4.3 Scope and Boundary of the Proof

The theoretical structure is intentionally limited. It uses zero-threshold sign binarization, jointly normal features, and large-sample covariance reasoning because those assumptions produce a clear analytical pathway. The dissertation does not claim that this proof is the final theory for arbitrary learned critical ranges. Instead, it uses the tractable result to establish a credible mechanism and then treats extension to nonzero and interval thresholds as part of the broader research agenda [20].

This bounded strategy is important to the dissertation’s argument. It allows the work to be rigorous without overclaiming. Because the theorem’s mechanism aligns with the empirical be-

havior observed in synthetic and real data, the case for the approach remains strong even though full generalization beyond the analyzed regime is still an open research direction.

4.5 HOW THE APPROACH MAPS TO THE RESEARCH QUESTIONS

The integrated pipeline naturally defines the dissertation’s three-question structure.

- **RQ1** evaluates whether the rectification-first pipeline improves sparse support concentration, lag localization, and practical usability relative to fitting on raw lag-expanded inputs.
- **RQ2** evaluates whether the theoretical mechanism behind those gains is credible, especially through IC-oriented arguments about correlation contraction and reduced inactive-feature interference.
- **RQ3** evaluates whether the sparse rectified model can be turned into a compact rule set without unacceptable operational loss.

Thus, the chapters that follow do not present three separate projects. They examine one pipeline from three necessary angles: empirical performance, theoretical justification, and deployment-ready interpretability.

CHAPTER 5

RECTIFICATION IMPROVES SPARSE LONGITUDINAL SELECTION (RQ1)

RQ1 asks whether critical-range rectification can produce a stable sparse baseline with reliable feature and lag attribution in longitudinal settings. This chapter answers that question by showing where and why rectification improves sparse longitudinal selection.

The evidence is anchored in prior work from the same method lineage, beginning with the rectification-first lasso logic framework and extending to later theory-backed case studies [20, 21]. Across studies, the central empirical pattern is consistent: when the data generating process has threshold-and-lag structure, rectification tends to improve support concentration, lag localization, and sparse-model usability under multicollinearity pressure.

5.1 A SIMPLE EXAMPLE TO AID INTUITION DEVELOPMENT

Consider a *toy example* showing how continuous features can obscure true relationships. Let $\mathbf{X}$ be a 10-feature matrix with columns $A, \dots, J$. Suppose only A and D determine the outcome y . Fitting LASSO on raw continuous data can yield spurious coefficients (Figure 2, top). After *binarizing* features by marking each as $+1$ or -1 based on a range, irrelevant columns disappear, and A and D stand out (Figure 2, bottom).

Binarization effectively “flattens” continuous variations that inflate feature correlations. Intuitively, columns with moderate overlap in continuous space can appear overly correlated. Mapping to $\{\pm 1\}$ reduces correlation unless columns are identical or scalar multiples, preserving meaningful relationships.

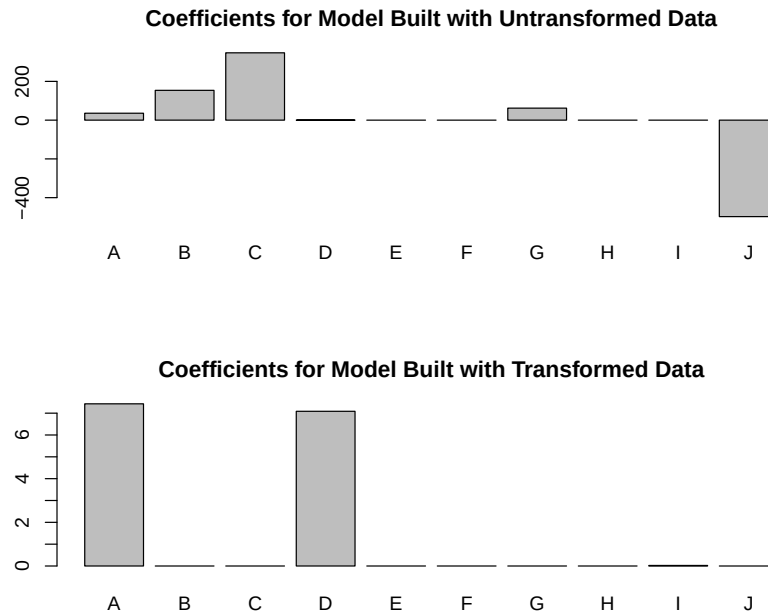


Figure 2. Coefficient vectors for LASSO on the same data before (top) and after (bottom) binarization. The non-zero feature coefficients indicate the selected features. These particular coefficient sets were calculated using the "glmnet" package from the R programming language, but similar results occur using LASSO implementations in other ecosystems.

5.1.1 Toy Example Data

The toy example data demonstrates the behavior of a simple set of instrument readings. A matrix with features 'A' through 'J' shows that even with only 10 features, the coefficients become muddled and tend to gravitate toward spurious correlations rather than the true support. Columns A and D were designed to produce the result vector, while the others were largely random, with top elements intentionally adjusted to ensure they are **not** optimal for producing the result vector.

$$\mathbf{X} = \begin{bmatrix} \text{A} & \text{B} & \text{C} & \text{D} & \text{E} & \text{F} & \text{G} & \text{H} & \text{I} & \text{J} \\ 0.49 & 0.48 & 0.42 & 0.13 & 0.45 & 0.48 & 0.42 & 0.48 & 0.43 & 0.46 \\ 0.12 & 0.52 & 0.46 & 0.47 & 0.42 & 0.47 & 0.49 & 0.47 & 0.51 & 0.48 \\ 0.23 & 0.11 & 0.38 & 0.57 & 0.99 & 0.51 & 0.41 & 0.71 & 0.29 & 0.56 \\ \mathbf{0.48} & 0.40 & 0.59 & \mathbf{0.46} & 0.41 & 0.43 & 0.40 & 0.41 & 0.60 & 0.52 \\ 0.83 & 0.45 & 0.48 & 0.54 & 0.44 & 0.44 & 0.23 & 0.41 & 0.46 & 0.49 \\ \mathbf{0.55} & 0.52 & 0.41 & \mathbf{0.58} & 0.40 & 0.60 & 0.51 & 0.42 & 0.42 & 0.45 \\ 0.53 & 0.02 & 0.55 & 0.16 & 0.38 & 0.06 & 0.50 & 0.55 & 0.71 & 0.51 \\ \mathbf{0.54} & 0.60 & 0.56 & \mathbf{0.47} & 0.49 & 0.52 & 0.52 & 0.56 & 0.54 & 0.58 \\ 0.33 & 0.22 & 0.59 & 0.49 & 0.47 & 0.49 & 0.59 & 0.47 & 0.91 & 0.57 \\ 0.34 & 0.49 & 0.66 & 0.51 & 0.48 & 0.79 & 0.84 & 0.53 & 0.73 & 0.67 \end{bmatrix}$$

5.1.2 Input Data Transformation

After transformation, the matrix is encoded with ones and negative ones. The encoding rule in this case was simple: values between 0.4 and 0.6 (inclusive) were encoded as ones; others were encoded as negative ones. Details regarding how to come up with this *critical range* are in the other sections. For now, a summarized version of the result is that if both features A and D were between 0.4 and 0.6 simultaneously, an event occurred.

$$\mathbf{X} = \begin{bmatrix} \text{A} & \text{B} & \text{C} & \text{D} & \text{E} & \text{F} & \text{G} & \text{H} & \text{I} & \text{J} \\ 1 & 1 & 1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 & 1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & -1 & -1 & 1 & 1 & -1 & 1 \\ \mathbf{1} & 1 & 1 & \mathbf{1} & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 1 & 1 & 1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 & 1 & -1 & -1 & 1 & -1 & -1 \end{bmatrix}$$

Here is the boolean response vector:

$$y^T = \begin{bmatrix} F & F & F & \mathbf{T} & F & \mathbf{T} & F & \mathbf{T} & F & F \end{bmatrix}$$

It can be shown that the only possible logical AND combination of these features which would produce the response vector shown would be the 1st and 4th columns.

Intuition Development

In the following examples, I show an easy way to visualize the logical process that helps develop an intuition for what types of problems this solution method works well for. A binary outcome is derived from a continuous feature measured over time. Events occur when the feature enters a specific range, shown as the shaded area in Figure 3. This behavior is common in physical phenomena that depend upon temperature or pressure, with phase changes being an obvious example. Collecting more data refines the event range. Even if only one relevant feature exists among many, defining binary values for all features highlights the causative feature. By replacing continuous measurements with binary values, the relevant features become more likely to have a larger

coefficient and the non-relevant features less. In the trivial case, if only one feature is relevant and fully describes the phenomenon, its binary vector will match the result vector.

Figure 3 illustrates a relevant and an irrelevant measurement. The orange curve (bottom) cannot be the sole cause since no clear relationship exists between its values and event occurrences. Applying this logic to all features, only the one defining all events without including non-events can be the sole cause.

The **critical range** (CR) of a feature is defined here as the range where specific events might occur based on prior behavior. The max and min values of the range correspond to feature values observed only during events. Feature combinations that fully explain events are included, while excluding non-events.

5.1.3 Expanding the Solution to Include Multiple Triggers

Generalizing this procedure to include situations in which an unknown number of features are required to simultaneously be within their own critical ranges for an event to occur is where the LASSO algorithm is useful.

Each feature's CR spans from the max value during an event to the min. Figure 4 shows two features where events occur only if both are in their CR.

The observer knows only the response vector showing when events occurred and the feature measurements across 17 time steps. In Figure 4, an event is possible when the top curve is in the upper shaded region and the bottom curve is in the lower shaded region. When both curves fall outside their respective ranges, the event probability is zero. Time steps 10 and 13 define the lower shaded region's boundaries for the bottom curve (similar logic for the top curve).

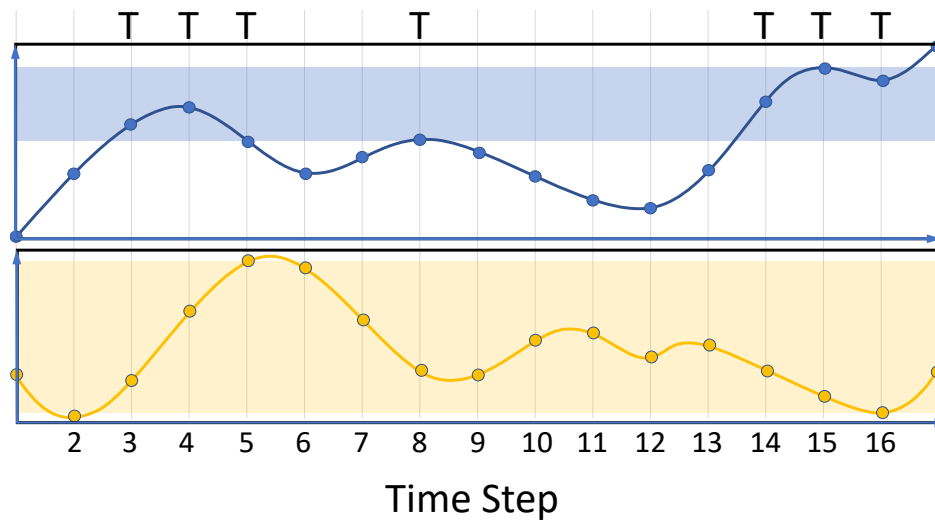


Figure 3. Measurement of two hypothetical features over sixteen time steps. Events are demarcated by the “T” over the curve, with the shaded area spanning the highest and lowest values at which the feature is measured during an event. Hypothetical ranges of interest are shown for an event. If the procedure for finding the critical range is performed on the orange curve (the critical range is shown by the orange shaded region on the bottom curve), it encompasses nearly the entire range of all measured values; this is an indicator that the feature represented by the orange curve is not relevant to event occurrence.

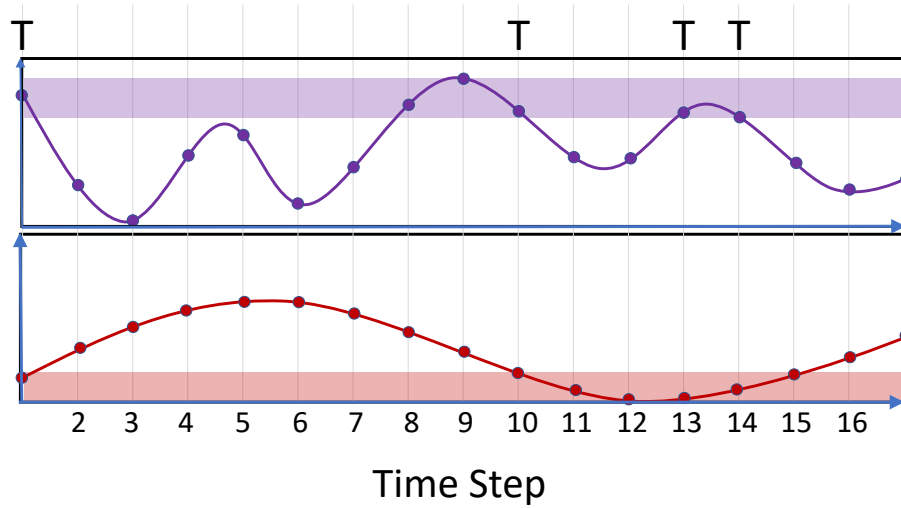


Figure 4. Hypothetical event occurrence in a more complex scenario. The occurrence of four events, each of which is labeled by the “T” symbol, is fully described by the two feature curves shown. For an event to occur, the top curve is required to be within the upper shaded region at the same time that the bottom curve is within the lower shaded region.

5.2 RQ1 EVALUATION FRAMEWORK

The dissertation evaluates RQ1 using three complementary evidence streams:

- **Controlled synthetic studies** with known feature-lag ground truth, where attribution quality can be inspected directly.
- **Comparative baseline studies** against methods designed to address collinearity or redundancy through alternative mechanisms.
- **Real-world case studies** that test whether observed gains survive outside synthetic conditions.

The evaluation criteria are practical rather than purely theoretical: discriminatory performance (for example AUC and Youden’s J), sparsity and support clarity, lag-level attribution fidelity, and end-to-end computational cost. This aligns with known limitations of penalty-only sparse methods under dependence, where prediction may remain strong while support recovery becomes unstable [5, 41]. The repeated-resample stability and stage-wise ablation package is summarized in Table 2 and Figures 6–9. The expanded baseline and cross-domain comparison package is summarized in Table 3 and Figures 14–18.

5.3 SYNTHETIC EVIDENCE WITH KNOWN GROUND TRUTH

Synthetic longitudinal datasets are constructed so that events occur when a small subset of variables simultaneously enters critical ranges at specific lags. This creates a hard but controlled setting where lag expansion induces severe multicollinearity by design, closely mirroring the failure modes discussed in lasso consistency literature [5, 41]. In this regime, representation-level rectification is applied before sparse fitting and compared with untransformed baselines.

The synthetic results in Table 1 show a consistent pattern: transformed models improve discrimination and often reduce runtime relative to their untransformed counterparts. More importantly for RQ1, support concentration is visibly improved at the coefficient level when transformed inputs are used, which directly affects lag-attribution reliability.

Figure 5 provides qualitative support for the same conclusion: rectification yields nonzero coefficients that align more closely with true causal lags, while untransformed fitting produces noisier support spread. This behavior is consistent with the rectification rationale described in prior publications, where binarized critical-range mapping is expected to reduce harmful correlation effects before sparse optimization [20, 21].

Table 1. Synthetic case: transformed vs untransformed comparisons from the dissertation experiments.

Method	Transform Status	Rel. Time	Youden's J	F1
LASSO	Transformed	1.00	0.975	0.988
LASSO	Untransformed	1.83	0.753	0.804
Random Forest	Transformed	12.5	0.980	0.990
Random Forest	Untransformed	43.0	0.913	0.947
Group LASSO	Transformed	24.5	0.689	0.558
Group LASSO	Untransformed	53.5	0.624	0.459

5.3.1 Repeated-Resample Stability and Stage-Wise Ablation

To test whether the synthetic gains persist beyond a single seeded draw, the dissertation adds a repeated-resample ablation on the Case 1 generator with fixed planted support and independent random seeds. Each resample is evaluated at three pipeline stages: (i) raw lag-expanded L1-logistic on the continuous design, (ii) rectified L1-logistic after training-only critical-range estimation, and (iii) a compact m -of- K rule derived from the rectified model by coefficient-ranked prefix compression. This design separates the contribution of the representation change from the contribution of the final logic-compression step.

Table 2 and Figure 6 show that rectification changes the result qualitatively. The gain is not marginal. Relative to raw sparse fitting, the rectified model improves mean test AUC from 0.986 to 0.998, improves Youden's J from 0.882 to 0.971, improves exact-lag F1 from 0.048 to 0.619,

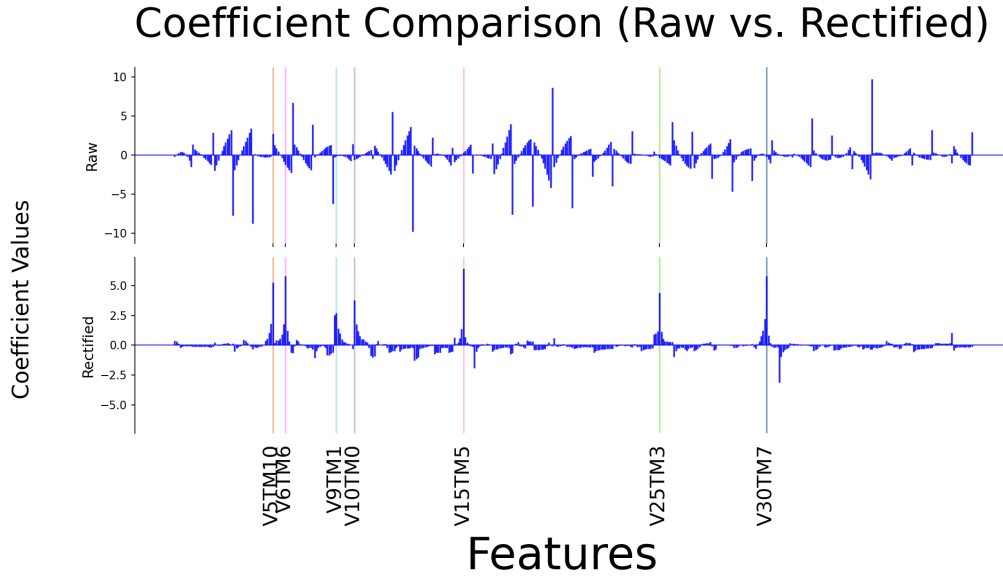


Figure 5. Coefficient profiles on synthetic data: transformed vs untransformed (ground-truth lag locations highlighted).

and reduces the average number of nonzero coefficients from 340.7 to 190.0. The compression stage then preserves essentially all of the discrimination gain while further increasing exact-lag F1 to 0.759 and reducing the deployed support to an average of six terms. This is exactly the practical pattern sought in RQ1: better attribution with no meaningful loss of operational performance.

The stability figures make the same point from two different angles. Figure 7 shows that pairwise exact-support overlap rises from 0.056 for the raw baseline to 0.452 after rectification, while the compressed rule reaches 0.600. Figure 8 shows why this matters: raw fitting spreads selection mass broadly across the lag-expanded design, whereas rectified models repeatedly return to the same planted variable-lag neighborhoods. The support recovered after compression is even more concentrated because low-weight spillover terms are removed instead of being retained for marginal predictive gain.

Table 2. Repeated-resample stability and ablation summary for the synthetic Case 1 study. Values are mean $\pm$ standard deviation across three independent resamples. Pairwise stability is summarized with exact-support Jaccard over all seed pairs.

Method	AUC	J	Exact-lag F1	Pairwise Jaccard	Total Nonzero	Runtime (s)
Raw	0.986 ± 0.008	0.882 ± 0.027	0.048 ± 0.082	0.056 ± 0.096	340.667 ± 105.396	23.142 ± 0.815
Rectified	0.998 ± 0.002	0.971 ± 0.019	0.619 ± 0.082	0.452 ± 0.090	190.000 ± 117.796	12.196 ± 1.562
Rule	0.994 ± 0.003	0.974 ± 0.015	0.759 ± 0.056	0.600 ± 0.346	6.000 ± 3.464	12.201 ± 1.562

Figure 9 shows that the representation change does not impose a prohibitive computational tax. As the lag-expanded feature count grows from 440 to 840, the rectification step remains small (0.015 to 0.034 seconds), while the rectified sparse fit remains materially faster than the raw fit (1.19 vs. 2.24 seconds at 440 features, and 2.30 vs. 4.35 seconds at 840 features). This matters for RQ1 because a stable baseline must remain practical under repeated resampling, ablation, model selection, and deployment refresh cycles.

5.4 COMPARISON TO COMPETING FEATURE-SELECTION BASELINES

RQ1 comparisons also include methods intended to address redundancy or grouped dependence through different mechanisms. Group and block-structured sparse families remain important comparators because they are explicitly designed for correlated predictor settings [12, 16, 34, 39, 40]. Across these studies, representation-level rectification combined with standard lasso provides clearer lag attribution in threshold-triggered synthetic settings.

A focused comparator is the quadratic-programming selector of Katrutsa and Strijov, which

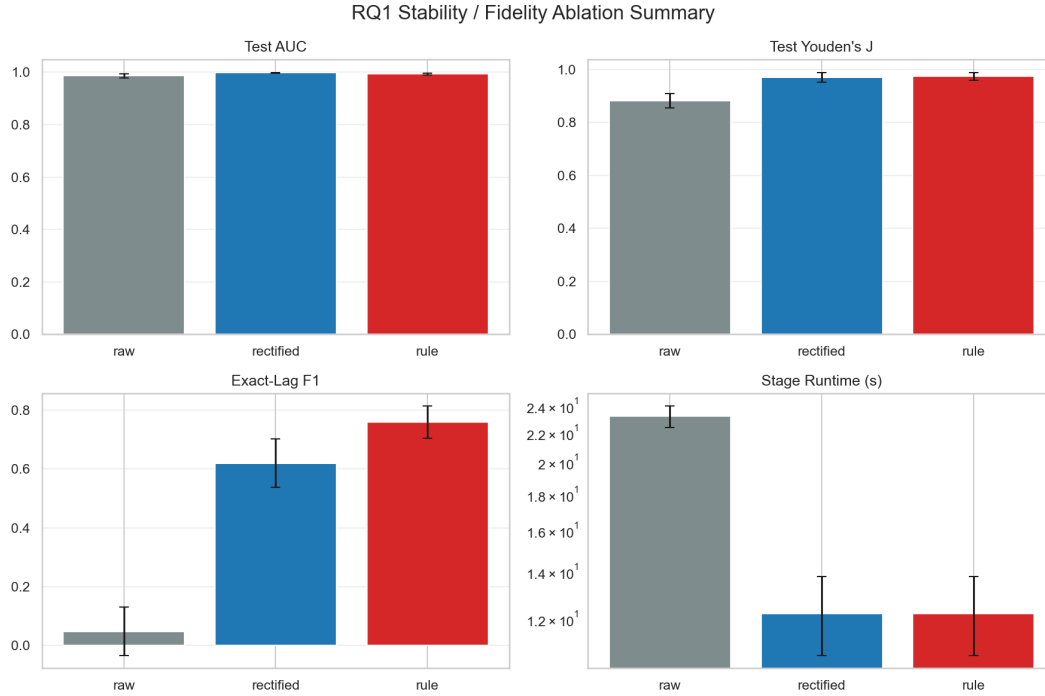


Figure 6. Repeated-resample ablation summary for the synthetic Case 1 study. Rectification improves discrimination and exact-lag recovery over raw sparse fitting, while the compressed rule retains nearly the same discrimination with stronger lag fidelity.

optimizes a relevance-redundancy objective using pairwise similarity structure [11]. The method is conceptually aligned with the goal of reducing redundancy, but it emphasizes pairwise criteria and does not directly encode multi-lag conjunctive trigger logic.

In the benchmark comparisons, the quadratic-programming approach often identifies related variables but may miss the correct lag localization and exhibits substantially higher computational cost. Observed runtime differences are on the order of roughly 100–200x relative to rectification plus lasso in the tested configuration. For RQ1, this matters because a "stable sparse baseline" is not only a statistical target; it must also be practical to run repeatedly across folds, ablations, and

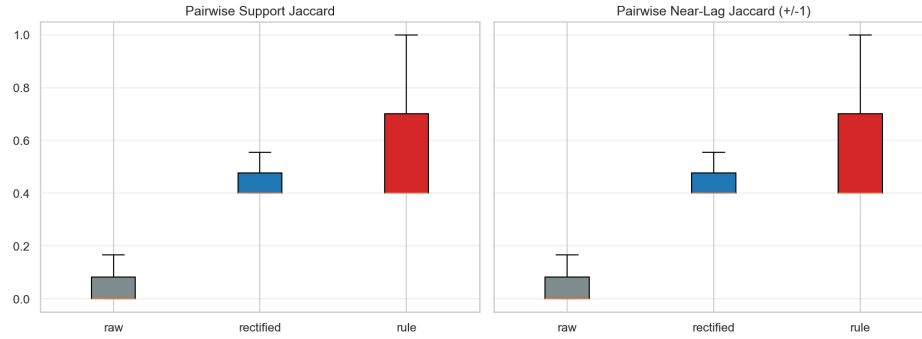


Figure 7. Pairwise exact-support and near-lag support Jaccard distributions across repeated resamples. Rectification produces a much more stable selected support than raw sparse fitting, and the compressed rule concentrates support most aggressively.

deployment refresh cycles.

5.4.1 Expanded Correlated-Feature Baseline Package

To broaden the comparator set beyond a single redundancy-oriented method, the dissertation adds a consolidated synthetic benchmark covering raw lasso, elastic net, adaptive lasso, group lasso, ordered-prefix selection, and the Katrutsa–Strijov quadratic-programming selector, alongside the rectified lasso baseline. All methods are evaluated on the same lag-expanded synthetic case used to stress attribution under multicollinearity.

Table 3 sharpens the baseline story. Raw lasso, elastic net, and group lasso all remain reasonably predictive, but none of them recover the planted lags exactly. Their support stays concentrated on correlated surrogates in the original continuous representation rather than on the true trigger lags. Adaptive lasso, ordered-prefix selection, and QP+-L1 reduce model size more aggressively, but they do so by pruning within the same raw representation and therefore sacrifice discrimination

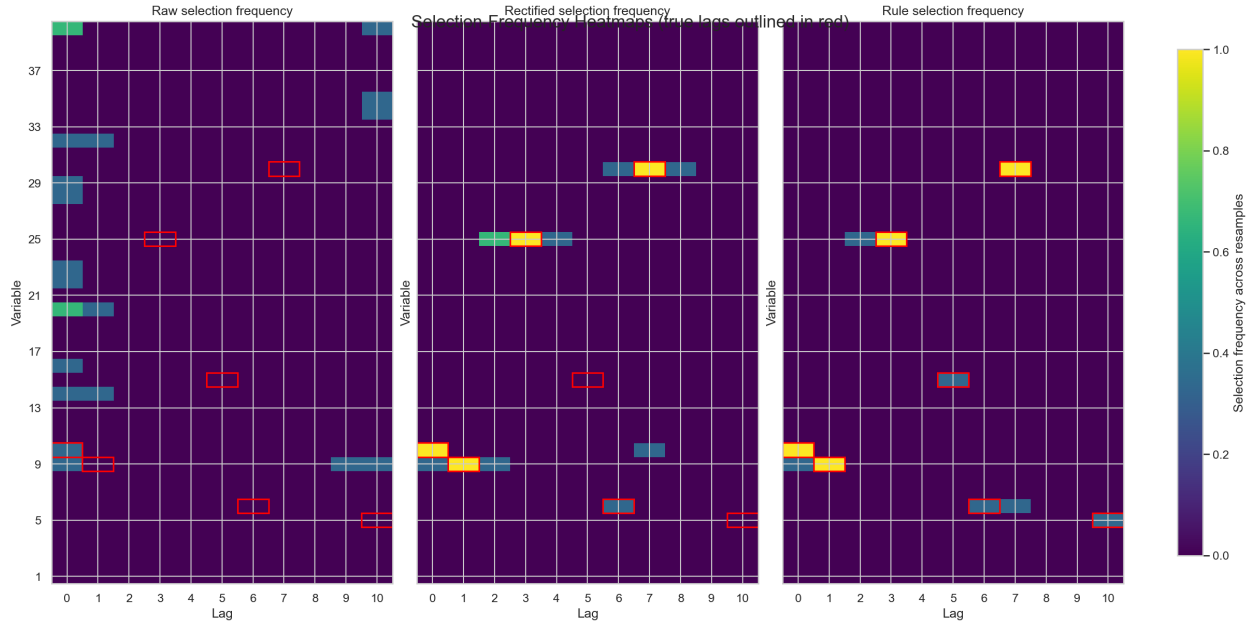


Figure 8. Selection-frequency heatmaps across repeated resamples. Red boxes mark the planted variable-lag pairs. Raw fitting diffuses support over unrelated variables and lags, whereas rectified and compressed models concentrate mass near the planted structure.

without solving the attribution problem. The rectified baseline is the only method in this expanded package that simultaneously attains the best AUC, the best Youden’s J , and nonzero exact-lag recovery. For RQ1, this matters more than a marginal runtime win because the question is not merely whether a sparse model predicts well; it is whether the selected support remains interpretable at the feature-lag level.

5.5 REAL-WORLD EVIDENCE AND INTERPRETABILITY TRADEOFFS

Synthetic evidence is necessary but not sufficient. Real-world studies therefore evaluate whether the same pattern appears in operationally relevant longitudinal data. The HAI industrial control benchmark is particularly useful because it includes realistic multi-sensor temporal dynamics with

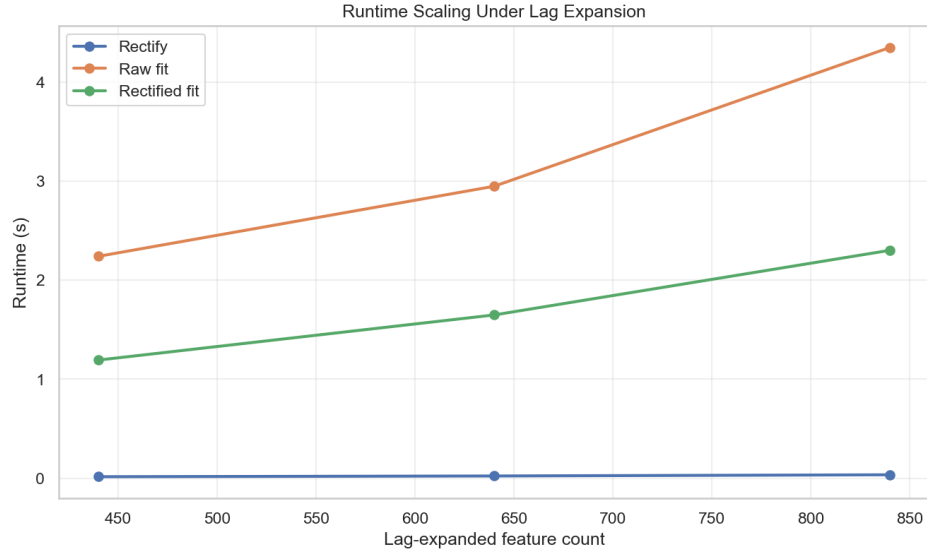


Figure 9. Runtime scaling as the lag-expanded feature count grows. The rectification step remains inexpensive, and the rectified sparse fit remains materially faster than the raw fit across the tested lag windows.

labeled attack scenarios and known subsystem context [27]. Additional evidence from historical ionospheric radar work supports the relevance of lagged signal discrimination settings where sparse attribution can complement raw predictive performance [28].

As shown in Table 4, HAI results favor transformed models on both discrimination and attribution-oriented interpretation. In this setting, transformed models concentrate coefficients on turbine-related channels aligned with known attack context, while untransformed models spread attribution more diffusely.

One subtle but important point in the HAI case is that residual nonzero coefficients in the $P4$ block should not automatically be interpreted as attribution failure. In HAI 1.0, $P4$ is the hardware-in-the-loop simulator that explicitly couples the boiler, turbine, and water-treatment subsystems

Importance Magnitudes for Quadratic Programming (Python qp-feature-selection, raw data)

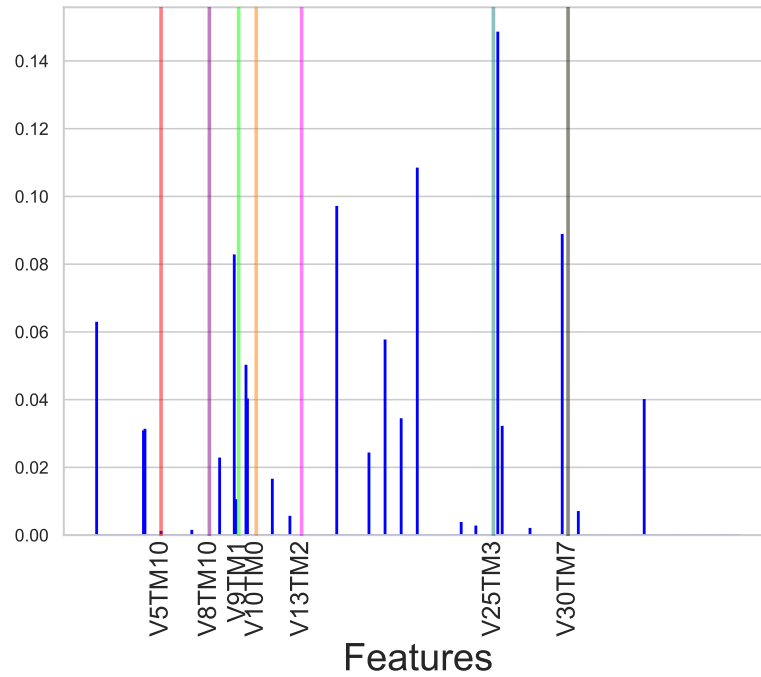


Figure 10. Quadratic-programming importance magnitudes on the synthetic dataset.

at the signal level, so a turbine-focused attack can legitimately induce secondary responses in simulator steam-side variables rather than remaining perfectly isolated within $P2$ [27]. For that reason, a transformed sparse model whose largest concentration remains in $P2$ but which retains a smaller, coherent set of $P4$ terms is more plausibly reflecting real cross-process dynamics than exposing a solver bug. In this benchmark, the appropriate attribution standard is therefore dominant concentration on the attacked subsystem with only limited, physically interpretable spillover into the coupled simulator block, not literal exclusivity of every nonzero coefficient.

Cross-dataset behavior is not uniform. In the UNICEF [36] case, untransformed models can achieve stronger raw discrimination metrics while transformed models remain substantially sparser

Table 3. Expanded correlated-feature baseline benchmark on the synthetic Case 1 study.

Exact-lag F1 evaluates whether the selected support lands on the planted variable-lag pairs rather than merely nearby correlated surrogates.

Method	AUC	Youden’s J	Exact-lag F1	Nonzero	Runtime (s)
Rectified L1	0.997	0.946	0.571	344	2.947
Raw L1	0.995	0.892	0.000	409	4.755
Raw Elastic Net	0.991	0.888	0.000	364	5.176
Raw Adaptive L1	0.982	0.835	0.000	53	2.110
Raw Group Lasso	0.990	0.888	0.000	385	1.488
Ordered Prefix	0.982	0.786	0.000	30	0.912
QP + L1	0.894	0.603	0.000	15	2.843

and easier to interpret. This pattern is consistent with prior reports that rectification benefits are context dependent: strongest in threshold-and-lag aligned regimes, and more mixed when data-generating structure is less compatible with the assumed critical-range mechanism [20].

5.5.1 Cross-Domain Transfer and the Goose Bay Protocol Audit

The broader transfer package applies one standardized generic protocol across four HAI attack subsets and the UCI ionosphere radar benchmark, then compares raw and rectified lasso under the same train/test handling and fixed penalty settings. Figure 14 shows that the transfer pattern is

Table 4. Real-world case studies: transformed vs untransformed comparisons from the dissertation experiments.

Statistic	HAI	HAI	UNICEF	UNICEF
	(Transformed)	(Un-Trans)	(Transformed)	(Un-Trans)
ACC	0.987	0.939	0.902	0.963
AUC	0.982	0.943	0.865	0.989
F1	0.987	0.942	0.894	0.958
Youden’s J-Index	0.974	0.878	0.815	0.925

mixed rather than uniformly positive. Rectification clearly improves both AUC and Youden’s J on the HAI turbine-loop attack, improves J substantially on the combined $P1/P2$ attack, is essentially neutral on one attack subset, and loses ground on another. The ionosphere row is also negative under this standardized protocol, with $\Delta J = -0.059$. This is informative rather than contradictory: it means the RQ1 claim must remain conditional on alignment between the rectification rule and the structure of the underlying domain.

The ionosphere result required a separate audit because, on its face, it appears to conflict with the positive Goose Bay result reported in the original case study [21]. Figure 15 resolves that apparent contradiction by keeping the two views distinct. The published 2022 Goose Bay reference moves from $J = 0.695$ to $J = 0.854$ after rectification, a gain of 0.159. The standardized generic transfer probe instead moves from $J = 0.775$ to $J = 0.716$, a loss of 0.059. These are not the same experiment. The published result used the original study protocol, while the generic transfer probe

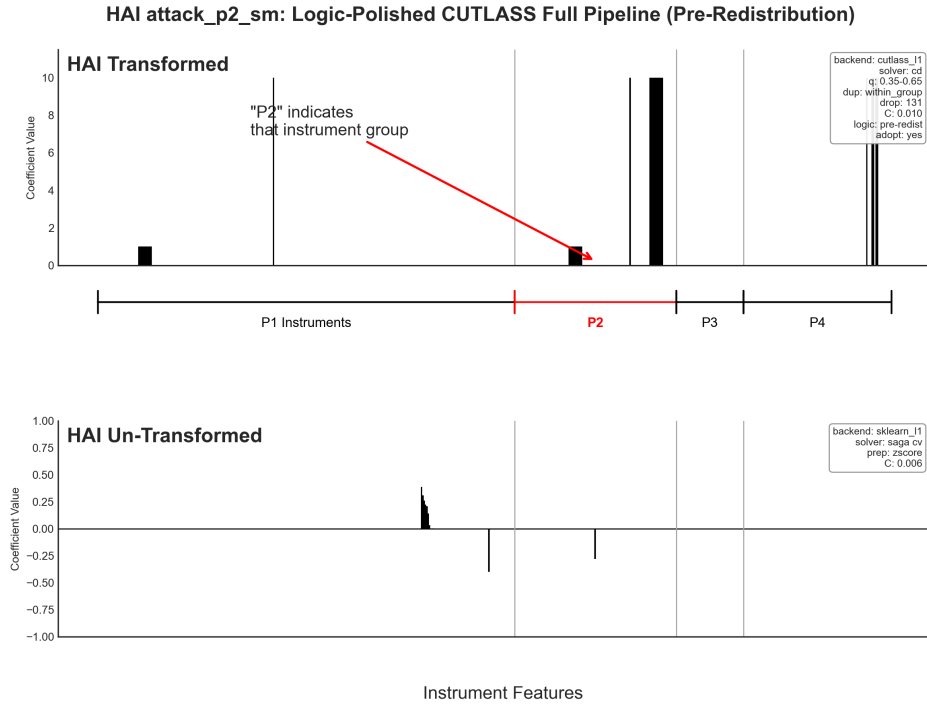


Figure 11. Real-world coefficient concentration and performance: HAI turbine-loop selection vs untransformed spread. Note that what is shown are the results after the "logic polishing" stage. Note the concentration of coefficients in the P2 region, with only one spike in the P1 region. The P4 region shows some response, but it is the software component of the HIL simulator and contains elements which tie into to the Turbine (P2) response.

deliberately used the same portable defaults as the other cross-domain datasets.

To determine why the Goose Bay views disagree, a repeated-split robustness audit was conducted that changes one protocol component at a time: train/test split, rectification rule, and penalty strength. Figure 16 shows that the split itself is not the main driver. Across 101 seeds, the mean rectified-minus-raw gain remains near zero when moving from the generic probe ($\Delta J = 0.006$) to a split-matched cutlass protocol ($\Delta J = 0.003$). The larger change appears when the rectifier se-

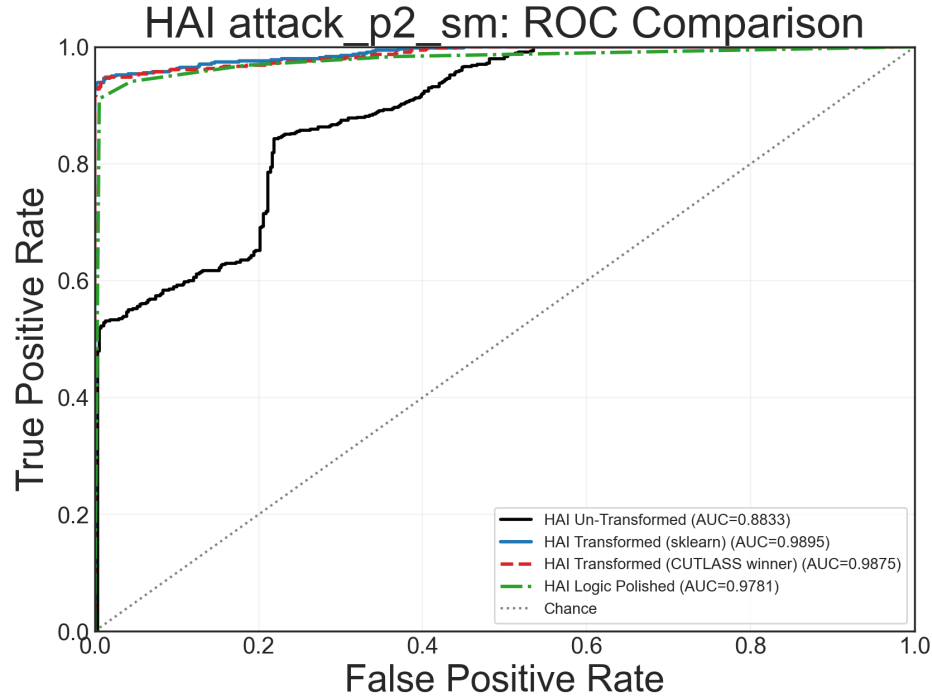


Figure 12. The Receiver Operating Characteristic (ROC) of several HAI analyses using different methods. While the logic polished version is slightly below the other transformed data curves in terms of the area under the curve (AUC), it is highly interpretable and is still superior to the model created from the un-transformed data.

mantics change from the default quantile rule to the positive-range min/max rule used in the earlier pipeline. Under the same 67/33 split and $C = 0.1$, the mean gain rises to 0.066 and is positive on 85.1% of seeds. Increasing the penalty parameter to $C = 10$ under the same min/max rectifier raises the mean gain further to 0.148 and makes it positive on every tested seed. On the original Goose Bay seed 2345, the progression is $-0.033 \rightarrow 0.065 \rightarrow 0.109 \rightarrow 0.210$, which again indicates that rectifier semantics and penalty strength matter more than split matching alone.

The protocol audit then motivates a direct sensitivity analysis of the two main tuning choices

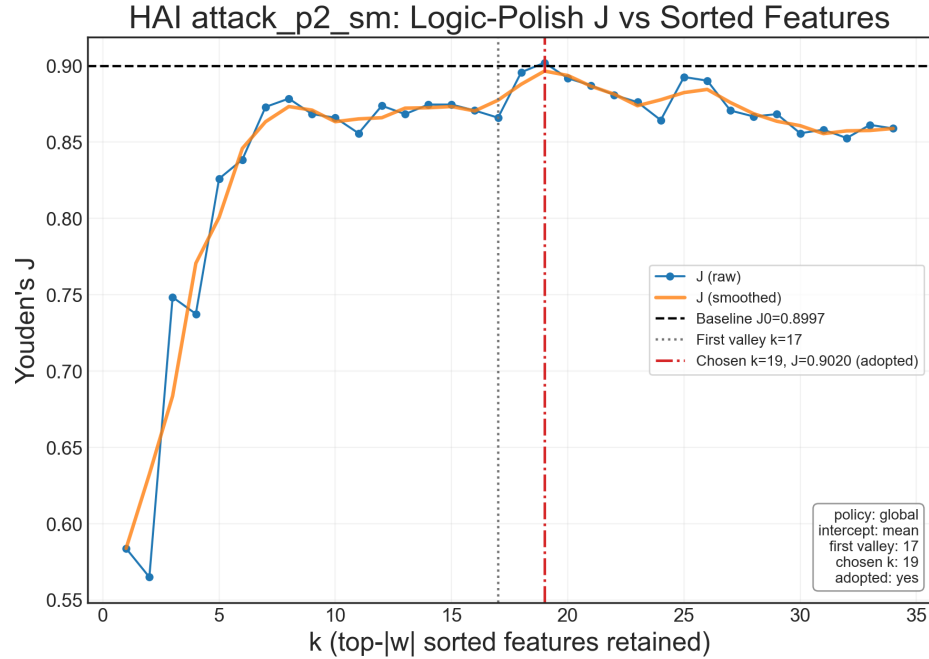


Figure 13. The Youden's J-Index is plotted against the features sorted by coefficient weight (pre logic-polish). Note that using a mere eight features achieves nearly as much at 0.88 as the maximum J-Index value of 0.9 at 19 features. Thus, the "anytime" nature of process reflects that a small group of highly relevant features can be used to create a simple logical rule which captures nearly all of the predictive power of the full model.

left in the standardized cutlass path: quantile bounds and penalty strength. Figure 17 shows that the default 25/75 quantile rectifier is close to neutral on Goose Bay ($\Delta J = 0.003$ on average), while wider windows improve steadily: 20/80 raises the mean to 0.063 and 10/90 to 0.082. Narrower windows move in the wrong direction, with 30/70 at -0.083 and 35/65 at -0.212 . The implication is not that one fixed quantile choice is universally optimal, but that the ionosphere case is unusually sensitive to the width of the positive-range window and behaves more favorably as the rule becomes more min/max-like.

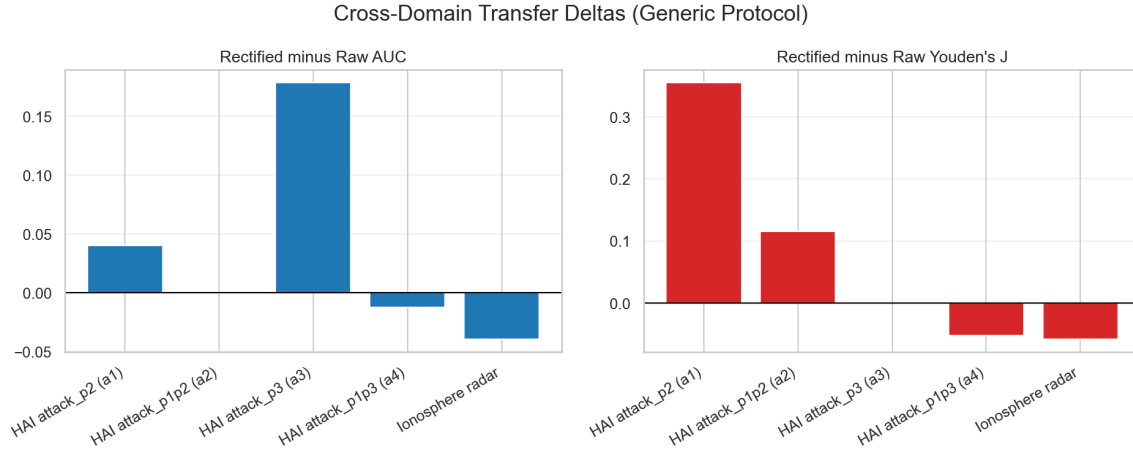


Figure 14. Cross-domain transfer deltas under one standardized generic protocol. Rectification improves some domains strongly, is neutral in others, and does not dominate uniformly across all settings.

Figure 18 extends the same idea to the lasso penalty. A log-uniform sweep over 100 values from 0.01 to 100 under the Goose Bay split and min/max rectification shows that the standardized choice near $C = 0.1$ sits well below the robust high- C plateau. Near $C = 0.108$, the mean gain is only 0.077. Over the broader 10–100 range, the mean gain stabilizes between roughly 0.149 and 0.157, and almost every seed is positive. There is also a narrow low- C sweet spot around $C \approx 0.025$ where the mean gain peaks near 0.200, but the lower standard-deviation panel shows that this region is materially less stable than the broader high- C plateau. The overlaid 10-fold cross-validation curve likewise flattens in the larger- C regime. For RQ1, the implication is practical: a standardized default such as $C = 0.1$ can understate the value of rectification on Goose Bay even though the domain still supports a strong rectified solution under a nearby and more stable operating region.

Taken together, the cross-domain and Goose Bay results refine the empirical claim rather than

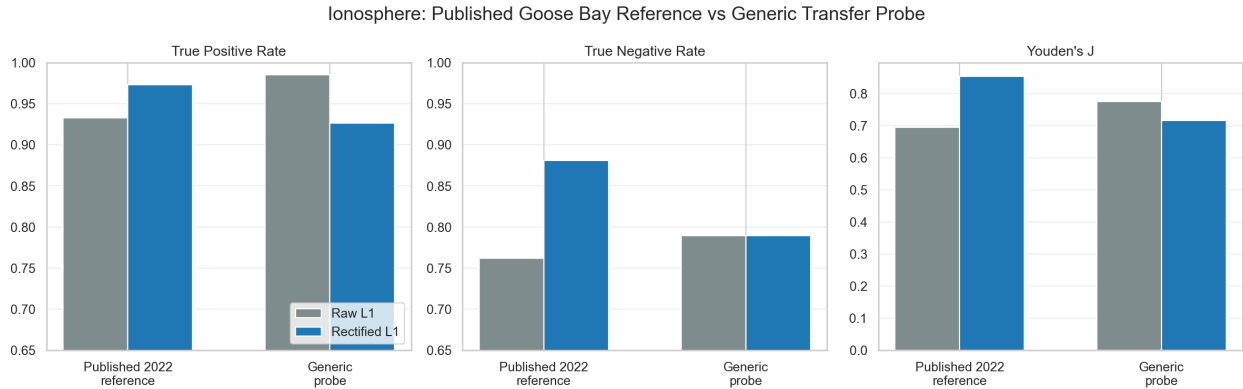


Figure 15. Ionosphere results separated into the published 2022 Goose Bay reference view and the standardized generic transfer probe. The negative generic-probe delta should therefore be read as a protocol-mismatch result, not as a replication of the original Goose Bay case study.

weakening it. The standardized transfer panel shows that rectification does not dominate uniformly across every dataset when one portable protocol is imposed. The Goose Bay audit then shows why one apparent counterexample appears: the ionosphere benchmark is highly sensitive to how the positive range is defined and to how strongly the lasso penalty suppresses correlated signal fragments. That is a useful scientific boundary. It lends credence to the claim that rectification-first sparse learning is strongest in threshold-mediated regimes when the rectification semantics are aligned with the event structure, not that one untuned default will be optimal for all situations. Further, these results are consistent with the hypothesis that, in threshold-mediated regimes, aligned rectification often improves the likelihood of more favorable irrepresentable-condition behavior. Empirically, once the rectifier definition and penalty are well matched to the domain, indicated by the cross-validation curve, rectification rarely hurts and usually helps.

5.6 ANSWER TO RQ1

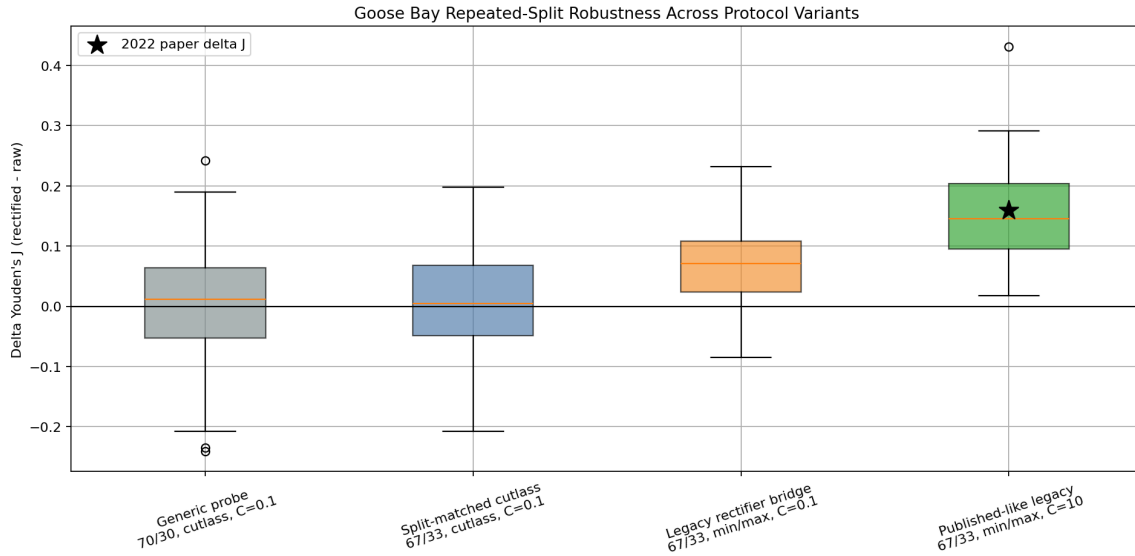


Figure 16. Repeated-split Goose Bay robustness across protocol variants. Matching the split alone does not recover the published gain; most of the movement comes from changing the rectification rule and then relaxing the penalty.

The answer to RQ1 is **yes, conditionally**. Critical-range rectification produces a more stable and interpretable sparse baseline in settings where events are driven by threshold-triggered lag interactions and raw lag expansion creates severe dependence. Evidence supporting this claim includes:

- improved lag-attribution concentration in controlled synthetic experiments,
- favorable or comparable discrimination with better sparsity in key real-world cases,
- and materially lower runtime than certain redundancy-oriented alternatives in tested configurations.

The repeated-resample stability and ablation results in Table 2 and Figures 6–9 reinforce this

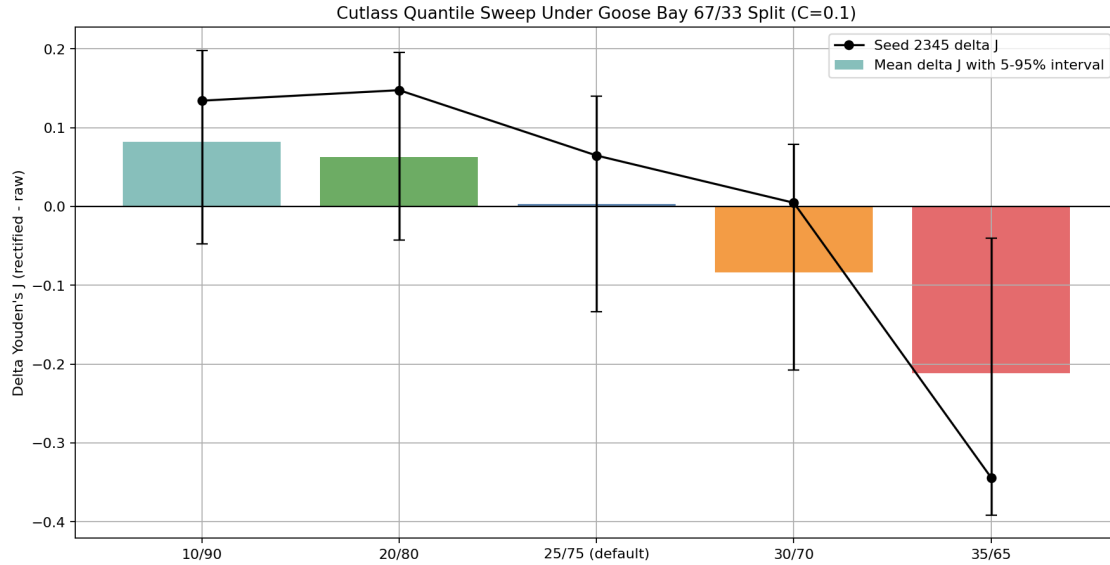


Figure 17. Sensitivity of Goose Bay performance to the quantile rectifier under the 67/33 split with $C = 0.1$. Wider positive-range windows improve performance relative to the default 25/75 setting, while narrower windows degrade it sharply.

conclusion, while Table 3 and Figures 14–18 show how the claim should be scoped. Expanded baseline coverage shows that penalty engineering alone does not recover exact lag attribution in the target synthetic regime. The cross-domain package shows that gains weaken or disappear when the data-generating process departs from threshold-mediated structure or when the rectification defaults are mismatched to the domain, as the UNICEF and generic Goose Bay probe illustrate. Even so, the dissertation establishes that representation-level rectification is a practical and effective intervention point for sparse longitudinal selection in the target regime.

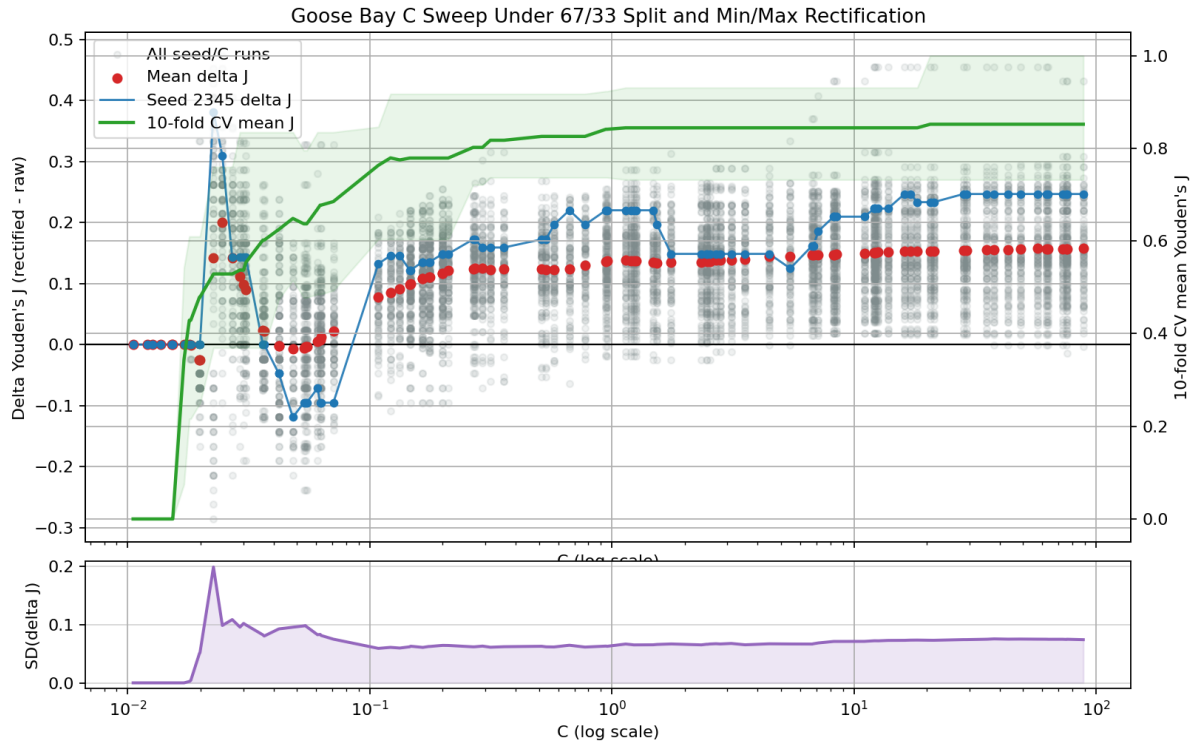


Figure 18. Goose Bay sensitivity to the lasso penalty under the 67/33 split with min/max rectification. The upper panel shows seed-level and mean ΔJ values with an overlaid 10-fold cross-validation mean- J curve; the lower panel shows the standard deviation of ΔJ at each C . The high- C regime is not uniquely optimal, but it is substantially more favorable and more stable than the standardized $C = 0.1$ setting.

CHAPTER 6

THEORETICAL ANALYSIS (RQ2)

RQ2 asks why rectification helps and how broadly that mechanism can be generalized. This chapter provides a tractable theoretical justification for the binarization step using the irrepressible condition (IC) as the central lens. The analysis is intentionally scoped: it focuses on zero-threshold sign binarization under joint normal feature assumptions, where correlation behavior is analytically tractable [20, 41].

The point of this chapter is not to claim a universal theorem for every thresholding scheme. Instead, it establishes a defensible mechanism: if binarization contracts dependence in the right way, the IC becomes easier to satisfy, and sparse support recovery becomes more likely. That mechanism aligns with both prior empirical evidence in this line of inquiry and broader observations that lasso-style selection can be unstable under strong dependence [5, 20].

6.1 WHY RQ2 MATTERS AFTER RQ1 EVIDENCE

RQ1 showed that rectification improves support concentration and lag attribution in threshold-aligned settings. However, empirical success alone is insufficient for the central claim. Without a credible mechanism, improvements could be dismissed as dataset-specific tuning effects.

For sparse selection methods, this concern is especially important. Lasso is a strong predictive tool [33], but model-selection consistency depends on structural conditions that may fail under multicollinearity [41]. In other words, high AUC does not imply reliable feature recovery. RQ2 therefore asks whether rectification improves a known structural bottleneck rather than merely

shifting surface metrics.

6.2 IC IN COMPUTER-SCIENCE TERMS

The IC can be read as an interference constraint in a sparse-recovery system. Let S be the true active-feature set and S^c the inactive set (the complement). For lasso to recover the right support, inactive features should not be too well explained by active features after accounting for active-feature geometry. Formally, one common sufficient condition is:

$$\left\| \mathbf{X}_{S^c}^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S) \right\|_\infty < 1.$$

A CS-oriented interpretation is useful:

- $\mathbf{X}_S^\top \mathbf{X}_S$ is the active-feature interaction matrix.
- $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}$ is a de-mixing operator for that interaction.
- $\mathbf{X}_{S^c}^\top \mathbf{X}_S$ measures leakage from active features into each inactive feature.
- The ℓ_∞ bound asks for worst-case leakage to stay below a hard margin.

When this leakage margin is violated, inactive features can mimic active ones and enter the model, taking the place of true support in the model, causing feature and lag misattribution due to spurious correlation. This is a structural reason for false inclusions and support instability, consistent with dependency-focused reviews [5].

6.3 TRACTABLE SETUP AND ASSUMPTIONS

To keep the analysis explicit, this chapter uses the tractable zero-threshold setting from prior work [20].

Data and transformation. Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be standardized continuous features and $\tilde{\mathbf{X}} \in \{\pm 1\}^{n \times d}$ the sign-binarized version:

$$\tilde{x}_{ij} = \text{sign}(x_{ij}),$$

with threshold at zero. For derivations, the appendix sometimes uses an equivalent $\{0, 1\}$ encoding; after centering, it preserves the same correlation-ordering arguments up to constant scaling.

Assumptions.

- Continuous features are jointly normal after standardization.
- The support set S is fixed in the local analysis.
- Correlation summaries are interpreted through large-sample covariance approximations.
- The goal is comparative: raw versus binarized IC satisfaction probability.
- The zero-threshold is meaningful and retains the signal components relevant to the analyzed mechanism in the target regime (e.g. modeling a threshold-based phenomena).

Joint normality and zero-threshold meaningfulness are modeling assumptions introduced for tractable analysis; they are not empirically verified prerequisites for using the full pipeline.

These assumptions are restrictive by design. They allow closed-form dependence mapping and a clean theorem statement, but they are not presented as full generality. In finite samples, especially when d is large relative to n , population-covariance approximations may deviate from realized sample behavior, so the claims here are regime-scoped and probabilistic.

6.4 WHY SIGN BINARIZATION CAN REDUCE DEPENDENCE

Under bivariate normality, sign-binarized correlation is a deterministic function of raw Pearson correlation (ρ):

$$\tilde{\rho} = \frac{2}{\pi} \arcsin(\rho).$$

For $\rho \in (-1, 1)$, this mapping contracts magnitude in most of the interior, while preserving sign and endpoint behavior. Intuitively, sign mapping removes amplitude information and retains only direction, which can weaken linear coupling caused by shared scale variation. All binarized equivalent values will be annotated with a tilde (e.g. " $\tilde{\rho}$ " for the binarized Pearson correlation).

From an IC viewpoint, this contraction can help in two places:

- It can reduce cross-correlation between inactive and active features.
- It can improve conditioning behavior of the active Gram block in relevant regimes.

The second point should not be read as a universal monotone guarantee for every design. It is a regime-level effect supported in the analyzed setting and tied to matrix-conditioning arguments [9, 20].

6.5 LEMMA PATHWAY (WITH INTUITION)

This section summarizes the proof structure. Full derivations are moved to the appendix.

6.5.1 Lemma 1: Correlation Contraction After Binarization

$$\tilde{\rho} = \frac{2}{\pi} \arcsin(\rho).$$

Interpretation: Under the PoC assumptions, sign binarization induces a deterministic arcsine mapping from raw correlation to transformed correlation. In practical terms, the representation

keeps polarity while suppressing magnitude information that drives linear coupling.

6.5.2 Lemma 2: Controlled Inverse Behavior for Active Block

$$|\tilde{\rho}| \leq |\rho|.$$

Interpretation: The transformed correlation magnitude is bounded by the raw magnitude (with equality at endpoints). This bound is the key contraction property used downstream to control active-block inversion effects and reduce amplification risk in de-mixing.

6.5.3 Lemma 3: Lower Inactive-Active Leakage

$$|\text{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j})| \leq |\text{Cov}(x_{S^c i}, x_{S j})|.$$

Interpretation: For each inactive-active pair, binarization weakens covariance leakage (or leaves it unchanged in edge cases). In sparse-selection terms, inactive features become less able to mimic the active span.

6.5.4 Lemma 4: Easier IC Satisfaction

$$\|\tilde{G}^{-1}\|_{\infty} \leq \|G^{-1}\|_{\infty} \quad (\text{positive-correlation analyzed regime}).$$

Interpretation: The transformed active block requires no greater inverse amplification than the raw block in the target regime. Combined with Lemma 3, this yields a smaller IC interference term and therefore a higher chance of satisfying the IC margin.

6.6 MAIN THEOREM (PROBABILITY FORM)

Let

$$\Theta(\mathbf{X}) = \left\| \mathbf{X}_{S^c}^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S) \right\|_\infty,$$

and define $\Theta(\tilde{\mathbf{X}})$ analogously for binarized features. In the analyzed setting, the target inequality is

$$P(\Theta(\mathbf{X}) < 1) \leq P(\Theta(\tilde{\mathbf{X}}) < 1).$$

This theorem states a probabilistic improvement claim, not deterministic per-instance dominance. The claim is that transformed designs are at least as likely to lie in an IC-favorable region. Strict improvement is expected only in stylized regimes.

6.7 PROOF SKETCH

A concise proof sketch is given below; full derivations remain in the appendix.

Sketch. For each inactive feature index $i \in S^c$, define

$$\theta_i = \left| \mathbf{c}_i \mathbf{G}^{-1} \text{sign}(\boldsymbol{\beta}_S) \right|,$$

where $\mathbf{c}_i$ is its covariance vector with active features and $\mathbf{G}$ is the active Gram matrix.

In large-sample approximation, replace sample quantities by population correlation blocks. For raw and binarized designs this gives comparable expressions based on (ρ_{iS}, ρ_{SS}) and $(\tilde{\rho}_{iS}, \tilde{\rho}_{SS})$ respectively.

By Lemma 1, the transformed inactive-active block contracts in norm under the analyzed regime. By Lemma 2, inverse amplification from the active block is controlled. Using submul-

multiplicative norm bounds,

$$\theta_i^{\text{bin}} \lesssim \|\tilde{\rho}_{iS}\| \|\tilde{\rho}_{SS}^{-1}\| \|\text{sign}(\beta_S)\|,$$

with an analogous bound for θ_i in the raw domain. The transformed upper bound is tighter under the lemma conditions, which implies a weakly higher probability that each inactive-feature interference score stays below 1. Taking the maximum over $i \in S^c$ yields the stated IC-probability improvement. $\square$

6.8 PRACTICAL INTERPRETATION

The theorem can be read as a data-representation effect on sparse search geometry.

- **Before transformation:** lag-expanded features create dense dependency graphs. Multiple inactive nodes have high edge weight to active nodes, increasing support ambiguity.
- **After transformation:** edge weights are attenuated in the target regime, reducing ambiguity and lowering the number of near-tie candidates along the regularization path.

This interpretation connects theory to observed outcomes in earlier chapters: fewer false positives, clearer lag attribution, and higher support stability in threshold-and-lag aligned datasets [20, 21].

Importantly, the argument does *not* require modifying lasso internals. The solver, objective, and tuning workflow remain standard; the intervention is in representation. That separation is operationally useful because it preserves mature optimization tooling while changing the statistical geometry presented to the solver.

6.9 A SMALL INTERFERENCE EXAMPLE

Consider a simplified setting with one active feature x_1 and one inactive feature x_2 that is strongly correlated with x_1 . If the true signal is carried only by x_1 , lasso can still assign weight to x_2 when x_2 is highly predictable from x_1 . In IC language, this is a high-leakage case: x_2 sits close to the active span, so the inactive-feature margin shrinks.

Now apply sign binarization. In the PoC regime, the effective correlation between the two channels is contracted by the arcsin mapping. The inactive channel remains related to the active channel, but less linearly confounded in the space used by the sparse solver. This can move the system from a near-violation regime toward a satisfied-margin regime.

For an advanced CS reader, this is analogous to reducing feature aliasing in a compressed representation. The model family is unchanged, but representational overlap among candidate predictors is reduced enough that greedy path updates and KKT-driven screening are less likely to elevate spurious coordinates early in the path.

6.10 HOW THIS DIFFERS FROM PENALTY-ONLY FIXES

A natural question is why this chapter emphasizes representation rather than only stronger penalties. Existing work already proposes many penalty-level fixes for dependence, including elastic net, adaptive penalties, and grouped or ordered structures [34, 40, 43, 44]. These methods are important and remain part of the baseline set.

The RQ2 claim is narrower: even with the same downstream sparse learner, changing the representation can improve the structural conditions that drive support recovery. Penalty redesign and representation redesign are not mutually exclusive; they attack different layers of the pipeline.

From a systems perspective:

- Penalty-level methods change optimization geometry in coefficient space.

- Rectification changes statistical geometry in feature space before optimization.

This distinction matters for engineering reuse. If the mechanism is mainly representational, the approach can be plugged into mature sparse-learning stacks without custom solvers, while still improving support behavior in targeted dependence regimes.

6.11 WHAT THIS THEOREM DOES NOT CLAIM

To avoid overreach, the following non-claims are explicit:

- It does not prove universal improvement for arbitrary nonzero thresholds.
- It does not guarantee gains for every dataset or every correlation sign pattern.
- It does not replace empirical validation; it explains a mechanism under a tractable regime.

These caveats are consistent with prior reports that transformed and untransformed tradeoffs can vary by regime, with some datasets favoring raw discrimination while transformed models improve interpretability and attribution fidelity [5, 20].

6.12 EMPIRICAL BOUNDARY CHECKS BEYOND THE THEOREM

The closed-form theorem is intentionally conservative. The dissertation therefore supplements it with empirical boundary checks that test how the mechanism behaves when its simplifying assumptions are relaxed. The corresponding threshold-perturbation studies, correlation-pattern sweeps, and negative-correlation summaries are reported in «< deficiency D003 >>.

1. Correlation-pattern sweeps track how support stability changes as the design approaches known IC failure regimes.

2. Threshold-placement perturbation studies measure how robust rectification remains when practical critical ranges depart from the idealized zero-threshold setting.
3. Negative-correlation and complement-feature cases identify where correlation contraction weakens or becomes asymmetric.
4. Cross-dataset comparisons separate settings where the theorem’s mechanism predicts observed gains from those where the benefits are primarily interpretability-driven.

6.13 EMPIRICAL IMPLICATIONS OF RQ2

The theorem’s implications appear in the dissertation results in three recurring patterns:

1. Under controlled dependence stress, transformed models show stronger support stability than their raw counterparts «< deficiency D003 »>.
2. In threshold-aligned regimes, false-positive lag selection drops as inactive-feature leakage is reduced.
3. Failure cases cluster where threshold mismatch or adverse covariance geometry breaks the conditions that make correlation contraction most helpful.

These empirical patterns convert RQ2 from an abstract theorem into a concrete explanation for the behavior observed in RQ1 and RQ3.

6.14 PRACTICAL SIGNIFICANCE

For RQ2, the core result is a principled explanation for why a simple preprocessing step can improve sparse support recovery in a difficult longitudinal regime. The significance is twofold:

- **Scientific:** it links empirical improvements to established model-selection theory through IC-oriented analysis [41].
- **Engineering:** it supports a modular pipeline design where preprocessing changes correlation geometry while downstream sparse solvers remain standard and efficient.

In summary, the theorem does not end the theoretical story, but it establishes a coherent bridge from binarization to IC favorability. That bridge is sufficient to support the dissertation's mechanistic claim while also identifying the boundary conditions that matter for later extensions.

CHAPTER 7

ANYTIME RULE COMPRESSION (RQ3)

Research Question 3 asks whether a sparse, rectified linear model can be compressed into a compact rule form while preserving practical discrimination quality. In this dissertation, the target is not just sparsity in coefficient space, but operational interpretability: a model that can be read, explained, and validated as an explicit decision rule under realistic constraints. The anytime rule compression framework addresses this requirement by converting a fitted sparse model into an ordered sequence of candidate logic rules and allowing compression to stop as soon as performance is good enough for the application context [19, 20].

7.1 WHY RQ3 MATTERS AFTER RQ1 AND RQ2

RQ1 and RQ2 establish two foundations: first, binarization plus sparse learning can recover relevant structure more reliably in threshold-driven settings; second, the theoretical conditions for support recovery can improve when feature interactions are reframed through the binarized representation. RQ3 addresses the remaining translational gap: whether those improvements can be turned into human-auditable, low-complexity rules without giving back the predictive benefit [19, 20].

This is important for longitudinal and event-focused domains where users need a concrete trigger logic rather than a dense weight vector. In healthcare monitoring, safety surveillance, and maintenance settings, end users often need to know which few conditions jointly trigger escalation, and they need that logic in a form that is stable under repeated deployment [8, 27, 37]. In

interpretability terms, this chapter focuses on the transition from post hoc explanations to inherently interpretable rule structure [17, 25].

7.2 FROM RECTIFIED SPARSE MODELS TO RULE SPACE

Given a rectified design matrix and an L1-regularized logistic model, the baseline predictor can be written as

$$\hat{p}(y_i = 1 \mid \mathbf{x}_i) = \sigma \left(\beta_0 + \sum_{j=1}^p \beta_j \tilde{x}_{ij} \right), \quad (1)$$

where $\tilde{x}_{ij}$ is the binarized/rectified feature and β_j is sparse due to the L1 penalty [6, 31, 33]. Let the active set be $\mathcal{A} = \{j : \beta_j \neq 0\}$. RQ3 starts from this fitted model and asks whether we can replace the full weighted sum with a smaller rule that preserves operating performance.

The compression idea is to map the active coefficients to a shared magnitude (logic polishing), keep their signs, and evaluate truncated models along an ordered path. Ordering is by absolute effect size from the trained sparse model so that early steps prioritize the most influential terms [19].

7.3 ANYTIME COMPRESSION OBJECTIVE

The objective is to optimize a practical trade-off:

$$\max_{\text{rule complexity } C} J(C) \quad \text{subject to} \quad C \leq C_{\max}, \quad (2)$$

where J is Youden’s statistic ($J = \text{TPR} + \text{TNR} - 1$), and complexity is represented by the number of retained conditions k (and optionally the vote threshold m in an m -of- k rule) [19]. Rather than commit to a single fixed complexity in advance, the anytime procedure scans candidate compressions and can stop at any prefix as soon as the achieved J is within an application-defined tolerance

of the baseline.

This aligns with a deployment reality: in some environments, a 1–2% relative loss may be acceptable if complexity falls by an order of magnitude; in others, compression is only accepted at near-zero loss. The framework supports either case by explicitly parameterizing the tolerance [19].

7.4 ALGORITHMIC PIPELINE

The compression process can be summarized in four stages [19].

7.4.1 1. Train Baseline Sparse Rectified Model

Fit L1-regularized logistic regression on the rectified matrix and compute baseline operating metrics (AUC, J , sensitivity, specificity) at the chosen operating threshold. This baseline defines the reference quality floor for compression [6, 19, 31, 33].

7.4.2 2. Logic Polishing and Ordered Prefix Construction

Restrict to active features $\mathcal{A}$ and sort them by $|\beta_j|$ descending. For prefix length k , keep the top- k signed indicators and assign common magnitude K (or equivalently use signed votes). This yields candidate rule families of increasing complexity, each nested in the next [19].

7.4.3 3. Vote Threshold Optimization

For each prefix, evaluate an m -of- k decision threshold. Intuitively, m controls how many supportive conditions are required before predicting the positive class. Sweeping m gives a local complexity-performance frontier at fixed k ; sweeping k gives the global anytime frontier [19].

7.4.4 4. Adoption Rule with Relative Tolerance

Let J_{base} be the baseline score and $J(k, m)$ the compressed candidate. A candidate is accepted when

$$J(k, m) \geq (1 - \varepsilon) J_{\text{base}}, \quad (3)$$

for configured tolerance ε . Among accepted candidates, one can choose the smallest k (maximal simplification) or the best J under a complexity cap. This makes compression policy explicit instead of ad hoc [19].

7.5 WHY THE METHOD IS “ANYTIME”

The method is "anytime" in the algorithmic sense: each additional step (larger k or alternative m) refines quality, but every intermediate rule is already a valid deployable model. If computation must stop early, the current incumbent can still be used. If more budget is available, scanning continues to improve operating characteristics. This property is practical for iterative model governance and constrained environments [19].

7.6 EMPIRICAL RESULTS

Across the dissertation experiments, compression reduces model complexity by large factors while preserving discrimination within application-level tolerance policies. The full real-data anytime curves, equivalence-style comparisons, and adoption-policy tables are reported in «< deficiency D004 >>. The synthetic and HAI examples shown in this document illustrate the same pattern: substantial simplification is possible before operating performance degrades materially [19].

Figure 19 illustrates the central behavior: J is tracked as a function of retained rule size k , creating an explicit curve from highly compressed to minimally compressed models. The chapter-level answer to RQ3 is based on this curve, not on a single arbitrarily chosen point.

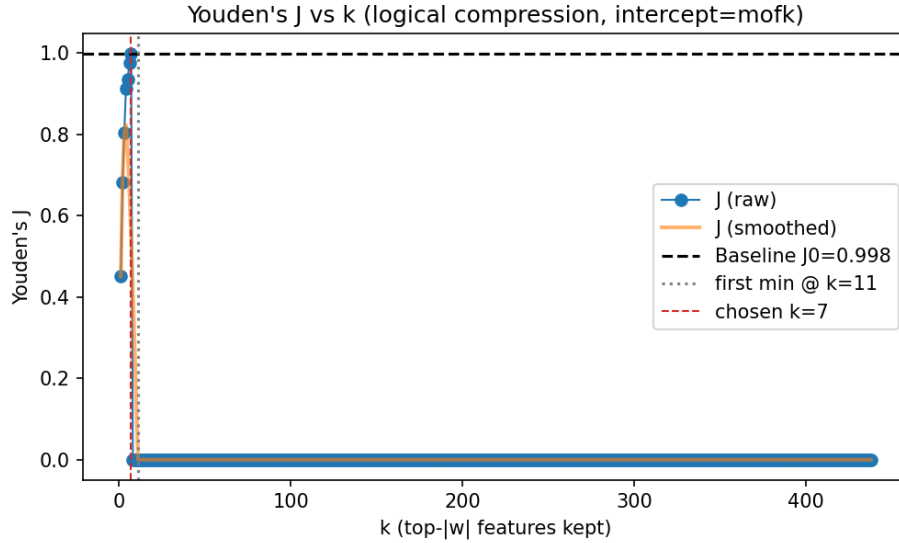


Figure 19. J versus retained rule size k for anytime compression. The curve provides a direct complexity-performance trade-off for selecting a deployable m -of- k rule. Note that the Youden's- J falls off to zero after $k=7$ because the algorithm exits early if it reaches a Youden's J -Index value of 1.0. See Figure 13 for an example of a real world dataset in which this is not the case.

7.7 STATISTICAL FRAMING OF “NO MATERIAL DEGRADATION”

RQ3 uses “no material degradation” in a practical, decision-oriented sense rather than a literal requirement of zero difference. Classical null-hypothesis difference tests are not sufficient to show retained utility, because failure to reject a difference is not evidence of equivalence. Accordingly,

the prior work used an equivalence framing (TOST-style logic) with a predeclared margin (for example, ± 0.01 AUC) to justify practical non-loss claims [19, 26, 30]. This is methodologically aligned with the chapter objective: demonstrate that simplified rules remain fit for purpose under explicit tolerance.

7.8 RELATIONSHIP TO ADJACENT INTERPRETABLE METHODS

Rule-list and scoring-system approaches provide strong interpretability baselines, but they often optimize directly in combinatorial rule space with different scalability profiles [1, 25]. The anytime compression strategy differs by leveraging a sparse convex fit first, then performing structured compression in a constrained post-processing stage. This separation is useful when the rectified feature space is large and sparse optimization is already part of the workflow.

The approach is also related to Logical Analysis of Data (LAD), particularly in its use of thresholded indicators and logical structure, but it targets a different optimization path and deployment interface [2, 3]. In the dissertation context, this can be viewed as a bridge between sparse statistical learning and logic-level decision rules tailored to threshold-driven events [20, 21].

7.9 MOTIVATION AND APPLICATION CONTEXT

The practical motivation for anytime rule compression is strongest in settings where decisions must be explained and audited quickly:

1. Clinical and public-health monitoring: produce concise trigger logic for intervention review and cross-site validation in longitudinal streams [8, 37].
2. Safety and reliability operations: support transparent condition-based escalation when false

negatives are costly and operators need explicit criteria.

3. Policy-facing analytics: convert technical models into rules stakeholders can inspect, challenge, and recalibrate without retraining a complex model stack.

Across these settings, the benefit is not only readability. Compact rules also reduce implementation burden, simplify drift monitoring, and support deterministic replay during audits. The anytime mechanism then provides a tunable point on the complexity-performance frontier for each deployment constraint [17, 19].

7.10 LIMITATIONS AND SCOPE

The results are strongest for the dissertation’s target regime: sparse, threshold-mediated phenomena represented through rectified features. This is a scoped framework, not a universal guarantee that every sparse model admits large compression at negligible loss. Performance depends on the shape of the learned coefficient spectrum, class balance, and operating-point priorities. These boundaries are consistent with the chapter’s scope and explain why compression is best treated as a policy-controlled operation rather than an unconditional simplification step.

7.11 ANSWER TO RQ3

The answer to RQ3 is **yes** within the dissertation’s target regime. Sparse rectified models can often be compressed into substantially smaller m -of- k rule representations with little practical degradation in discrimination, while improving interpretability and deployability. The full real-data compression frontiers and equivalence-style summaries reported in «< deficiency D004 »> reinforce this conclusion. The key contribution is the anytime formulation itself: it makes

compression controllable, auditable, and explicitly tied to operational tolerances rather than fixed model-size heuristics [19, 20].

CHAPTER 8

FUTURE WORK

This dissertation establishes a rectification-first pipeline for interpretable longitudinal modeling and shows how sparse fitting and anytime compression can be integrated into a single decision-support workflow. The most useful future work is therefore not to revisit the core pipeline, but to broaden its theoretical coverage, expand its empirical boundary, and deepen its deployment evidence.

8.1 EXTENDING THE THEORETICAL SCOPE

The theorem in Chapter 6 is intentionally centered on zero-threshold sign binarization under joint normality. A natural next step is to extend the analysis to learned nonzero critical ranges, finite-sample concentration bounds, and adverse covariance geometries. The most valuable outcome would be a family of regime-specific guarantees that clarifies when rectification helps because of dependence contraction and when other mechanisms dominate [20, 41].

8.2 BROADER EMPIRICAL VALIDATION

The dissertation’s strongest gains appear in threshold-mediated settings. Future work should therefore test the same pipeline on additional clinical, industrial, and socio-environmental longitudinal benchmarks with standardized stability reporting and prospective evaluation splits [8, 27, 37, 38]. Those studies would sharpen the applicability boundary by separating regimes where rectification improves both discrimination and attribution from regimes where its main benefit is

interpretability.

8.3 RICHER RULE FAMILIES AND HYBRID MODELS

Anytime compression currently targets compact m -of- K style rules. Future work should explore richer rule structures, including hierarchical rules, interaction-specific clauses, group-aware compression, and hybrid models that retain both threshold indicators and selected continuous magnitude terms. Such extensions could preserve the current pipeline’s transparency while reducing performance loss in settings where post-threshold magnitude still carries predictive information [1–3].

8.4 PROSPECTIVE DEPLOYMENT AND HUMAN EVALUATION

The framework is designed for auditable decision support, but deployment studies remain an important next step. Future work should evaluate how analysts, operators, or clinicians use the extracted rules in practice, whether compression improves review speed and consistency, and how rule updates should be governed under concept drift. Human-centered validation is especially important in high-stakes environments where explanation quality matters alongside discrimination [17, 25].

8.5 TOOLING AND REPRODUCIBILITY

Another useful direction is to package rectification, sparse fitting, and compression into a fully scripted artifact bundle with dataset manifests, versioned splits, and one-command regeneration of every figure and table. This would make the framework easier to audit, benchmark, and extend in later work, and would lower the cost of reproducing both positive and negative findings.

8.6 CLOSING PERSPECTIVE

Taken together, these directions move the dissertation from a strong pipeline result toward a more general methodology for interpretable longitudinal decision support. The core framework is now established; the next phase is to broaden its guarantees, domains, and deployment evidence.

CHAPTER 9

CONCLUSION

This dissertation examined longitudinal feature learning under three simultaneous constraints: dependence induced by lag expansion, the need for reliable feature-lag attribution, and the requirement for compact, human-auditable model outputs. The central claim is that a representation-first pipeline, followed by sparse fitting and anytime rule compression, provides a practical middle ground between unstable raw sparse selection and computationally heavy direct rule search [19–21].

9.1 RESEARCH PROGRAM SUMMARY

The work was organized around three research questions:

- **RQ1:** whether critical-range rectification improves sparse attribution reliability in longitudinal settings;
- **RQ2:** why rectification helps under a defensible theoretical mechanism and where that mechanism is scoped;
- **RQ3:** whether sparse rectified models can be compressed into compact logical rules without unacceptable operational loss.

Together, these questions move from empirical behavior (RQ1), to theoretical plausibility (RQ2), to deployment usability (RQ3). This sequencing is intentional: interpretability claims are weak if support recovery is unstable, and practical rule compression is less meaningful without

a credible upstream selection mechanism [5, 17, 25, 41].

9.2 CONCLUSIONS BY RESEARCH QUESTION

9.2.1 RQ1 Conclusion: Rectification Can Improve Attribution Reliability in Target Regimes

The accumulated evidence supports a conditional positive conclusion for RQ1. In threshold- and-lag aligned settings, critical-range rectification tends to improve support concentration, lag localization, and sparse-model usability under multicollinearity stress [20, 21]. This finding is consistent with known dependence-related limits of penalty-only sparse selection, where prediction can remain acceptable while support identification degrades [5, 41].

The dissertation does not claim universal dominance over all baseline families. Instead, it establishes that representation-level intervention is a viable and often effective alternative to penalty redesign alone, especially when events are naturally threshold-mediated.

9.2.2 RQ2 Conclusion: A Scoped IC-Based Mechanism is Theoretically Defensible

For RQ2, this work provides a scoped theoretical result: in a tractable regime (zero-threshold sign binarization with explicit assumptions), dependence contraction can improve the likelihood of satisfying IC-related sparse-recovery conditions [20, 41]. The theoretical contribution is therefore mechanistic rather than universal.

This distinction is central to rigor. The chapter argues for a plausible structural pathway from binarization to improved sparse-support behavior, while explicitly separating theorem-scoped claims from broader engineering intuition. The resulting conclusion is that rectification is not just a heuristic preprocessing trick; it has a coherent connection to established sparse-selection theory

under defined conditions.

9.2.3 RQ3 Conclusion: Anytime Compression Makes Interpretability Operational

For RQ3, this work concludes that sparse rectified models can often be compressed into substantially smaller rule representations while preserving practical discrimination quality under pre-defined tolerance policies [19, 20]. This directly addresses deployment requirements in settings where explanation, auditability, and compact decision logic are non-negotiable [17, 25].

The key advance is the anytime framing itself: rule complexity can be reduced incrementally with explicit stop/adoption criteria, making the interpretability-performance tradeoff controllable rather than ad hoc. Equivalence-oriented evaluation framing further supports practical non-loss claims when simplification is the objective [26, 30].

9.3 PRIMARY DISSERTATION CONTRIBUTIONS

The dissertation contributes the following integrated advances:

1. **A rectification-first longitudinal feature-learning pipeline** that preserves lag structure while mapping signals into critical-range indicators with clear operational semantics [20, 21].
2. **A scoped theoretical bridge to sparse-recovery conditions** showing how representation-level dependence contraction can increase IC favorability in a tractable regime [20, 41].
3. **An anytime rule-compression stage** that converts sparse coefficients into compact m -of- k -style logic with tunable performance-complexity tradeoffs [19].
4. **A unified evaluation perspective** that balances discrimination, attribution stability, inter-

pretability complexity, and computational efficiency rather than optimizing a single metric.

5. **A reproducible implementation pathway** for rectification, sparse fitting, and compression, enabling repeatable comparisons and practical extension in follow-on work.

9.4 POSITIONING RELATIVE TO PRIOR WORK

This work extends, rather than replaces, prior sparse and interpretable modeling literature. Relative to classical L1 and elastic-net families [6, 31, 33, 43, 44], the main difference is intervention point: representation is modified before sparse optimization. Relative to grouped and ordered penalties [16, 34, 35, 39, 40], this work emphasizes threshold logic and lag-attribution clarity in a longitudinal context. Relative to direct rule-learning approaches [1–3], it uses sparse convex fitting as a scalable front end and performs structured rule simplification afterward.

Accordingly, the work is best interpreted as a bridge architecture between sparse statistical learning and rule-level interpretability, not as a claim that one method family should replace all others.

9.5 SCOPE BOUNDARIES AND LIMITATIONS

Several limitations remain explicit:

- Benefits are strongest in regimes with threshold-mediated event structure and may be weaker when that structure is absent.
- The current theory is intentionally scoped and does not provide universal guarantees across arbitrary threshold schemes or dependence geometries.

- Compression quality depends on operating-point policy, class balance, and the shape of the learned sparse support.
- Interpretability gains can trade against raw discrimination in some datasets.

These boundaries are consistent with broader evidence that no sparse-selection or interpretable-modeling method is uniformly optimal across all dependence structures and objectives [5, 17].

9.6 PRACTICAL IMPLICATIONS

This work’s practical implication is straightforward: when longitudinal decisions require both predictive utility and audit-ready rationale, it can be more effective to restructure features around event-relevant critical ranges first, then apply mature sparse optimization, and finally compress to explicit rule logic. This design is particularly relevant in domains such as ICS monitoring and clinical trajectory analysis, where lag effects, threshold triggers, and traceable decision criteria are operationally important [8, 27, 37].

9.7 FINAL STATEMENT

This work concludes that rectification-first sparse longitudinal learning with anytime rule compression is a credible and useful framework for interpretable event modeling under dependence. Its value lies in integration: empirical support behavior (RQ1), scoped mechanistic theory (RQ2), and operational rule simplification (RQ3) are addressed as a single pipeline rather than isolated techniques [19–21]. Within its stated scope, this provides a defensible contribution to interpretable machine learning for longitudinal data.

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APPENDIX A

SYNTHETIC DATA DEVELOPMENT

The following process is common to all of the synthetic data sets.

The simulated data was generated with the following requirements in mind in order to simulate a time-varying sensor reading:

- The data needed to be time-varying. Each independent variable had a continuous stream of data that varied over time. This was necessary to generate realistic scenarios that show the potential for analyzing lagged longitudinal data.
- The data needed to be range bound. This reflected the desire to model realistic physical attributes, which tend to neither rise nor fall without end.
- The data needed to be complex. In order to provide a convincing test, it needed to have no easily discernable pattern.
- The data needed to be cyclic. Since the idea was to simulate sensor data or physical processes, and enough interactions needed to be generated in order to create a robust data set, some sort of cyclic process was necessary.

To meet these requirements, a sine function was chosen as the basic building block for the simulated data. Some additional details:

- When necessary, the Mersenne-Twister pseudorandom algorithm was used to generate pseudorandom numbers. This is the default algorithm when generating pseudorandom numbers via the base R package.

- To generate complexity, multiple sine waves of varying frequency and amplitude were combined via superposition.
- The data was generated first, and then particular data streams were chosen and designated as the “relevant” features.

The procedure:

1. Generate a large number of sine waves with five possible randomly chosen frequencies per cycle. Something like the following curve is generated:

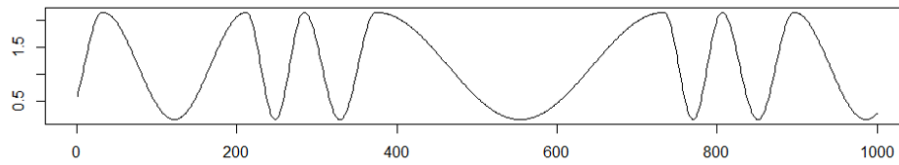


Figure 20. The first step in the synthetic data curve generation process.

2. Superpose several of the curves generated in step 1. Something like the next curve is generated by this step:

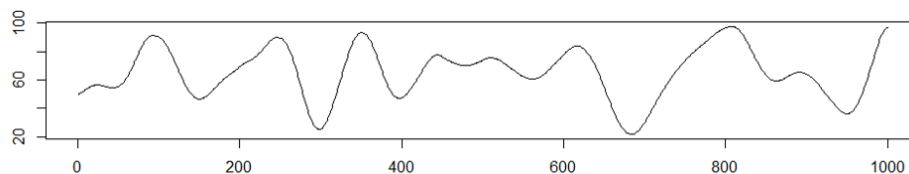


Figure 21. The second step in the synthetic data curve generation process.

3. To create a diversity of values a new range was selected using a pseudorandom normal distribution centered on zero, standard deviation of 0.05, and multiplied by 100. This produced a large enough diversity of magnitudes to demonstrate the range agnostic quality of the method.

4. A version of the waves was shifted by a pre-determined amount to simulate lag. The next graphic shows an example of this. Using lagged data simulates sensor data with readings that are time-late by an unknown amount, or an effect that has an unknown delay time between the change in reading and the occurrence of an event.

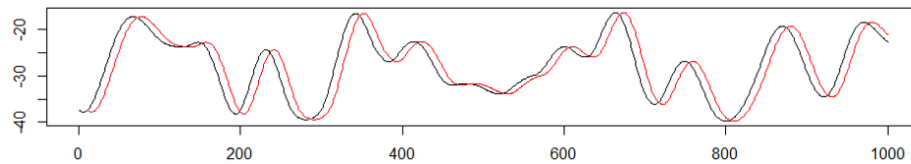


Figure 22. The fourth step in the synthetic data curve generation process.

5. Assemble enough curves to have a large selection of potentially “relevant” and non-relevant curves. Figure 1 shows how the relevant curves are interpreted to generate events. Note that the shifted curves determine when the event occurs, while the unshifted curves are matched with the time step in which they occurred to compose the data.

6. Each time step is interpreted as a separate example, and if all relevant curves meet the selection criteria during that time step, it is labeled as a “True” example, and otherwise is labeled as “False”. The data for each example is taken from the unshifted curves.

7. The data is next flattened such that a row contains the data from the current time step as well as the ten time steps previous (this is adjustable in the script). The data from time steps zero (the

current time step) through nine (the data nine time steps in the past) on the row for one example, for instance, will be the data for time steps one through ten on the next chronological example.

The next graphic illustrates:

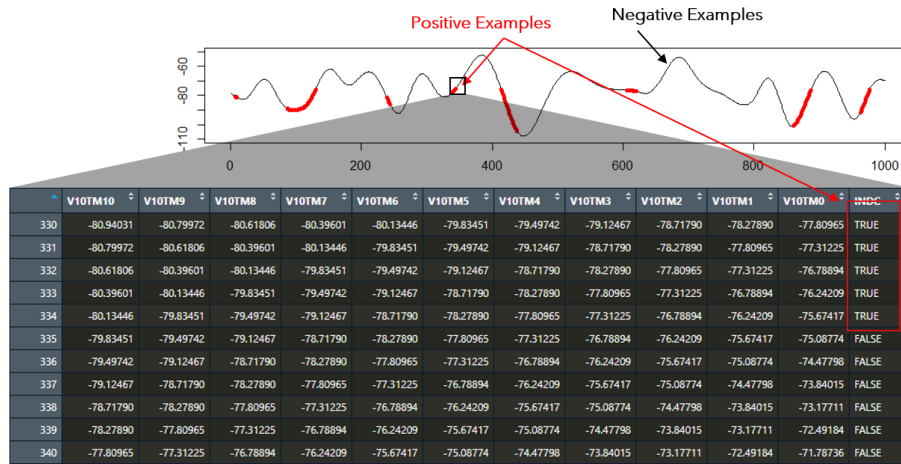


Figure 23. The seventh step in the synthetic data curve generation process, flattening the curve.

8. The final step is to randomize the rows to prevent any relationship between time steps from influencing the final outcome of the experiment. There is no indication in the flattened data features where an example belongs chronologically with respect to the other examples.

9. Once the data set has been created, it is split into training and testing sets with 70% of the data used for training and 30% of the data used for testing.

The procedure described generated a data set that included several relevant features combined via a logical AND, is representative of what might be observed with real sensor measurements and is complex enough to prevent any latent patterns within the data from influencing the results.

APPENDIX B

LEMMAS

B.1 LEMMA 1

The relationship between the pre-binarization correlation (ρ), and the post binarization correlation ($\tilde{\rho}$) for **standard normal** random variables is:

$$\tilde{\rho} = \frac{2}{\pi} \arcsin(\rho)$$

Step 1. *Basic definitions.*

Let X and Y be **standard normal** random variables with:

- **Means:** $E[X] = E[Y] = 0$ (centered)
- **Variances:** $\text{Var}(X) = \text{Var}(Y) = 1$ (standardized)
- **Correlation:** $\text{Corr}(X, Y) = \rho$, where $-1 \leq \rho \leq 1$

Step 2. *Define the Binarized Variables.*

We binarize X and Y using the threshold zero:

$$\tilde{X} = \begin{cases} 1 & \text{if } X > 0 \\ 0 & \text{if } X \leq 0 \end{cases} \quad \text{and} \quad \tilde{Y} = \begin{cases} 1 & \text{if } Y > 0 \\ 0 & \text{if } Y \leq 0 \end{cases}$$

This 0/1 encoding is affine-equivalent to the $\{\pm 1\}$ sign encoding used in Chapter 06; after centering, the correlation-ordering arguments are unchanged up to constant scaling.

Step 3. *Calculate Means and Variances of Binarized Variables.*

Since X and Y are symmetric about zero:

- Mean of $\tilde{X}$: $E[\tilde{X}] = P(X > 0) = 0.5$
- Variance of $\tilde{X}$: $\text{Var}(\tilde{X}) = E[\tilde{X}^2] - (E[\tilde{X}])^2 = 0.5 - (0.5)^2 = 0.25$
- Similarly for $\tilde{Y}$: $E[\tilde{Y}] = 0.5$, $\text{Var}(\tilde{Y}) = 0.25$

Step 4. *Express the Correlation Between Binarized Variables.*

The correlation $\tilde{\rho}$ between $\tilde{X}$ and $\tilde{Y}$ is:

$$\tilde{\rho} = \frac{E[\tilde{X}\tilde{Y}] - E[\tilde{X}]E[\tilde{Y}]}{\sqrt{\text{Var}(\tilde{X})\text{Var}(\tilde{Y})}} = 4(P(\tilde{X} = 1, \tilde{Y} = 1) - 0.25)$$

Step 5. *Define the probabilities.*

Since X and Y are assumed to be jointly normally distributed, $P(\tilde{X} = 1, \tilde{Y} = 1)$ can be expressed using the **bivariate standard normal cumulative distribution function (CDF)**.

$$P(\tilde{X} = 1, \tilde{Y} = 1) = P(X > 0, Y > 0)$$

Step 6. *Use the Bivariate Normal CDF.*

For standard normal variables X and Y with correlation ρ an expression of the *Gaussian Copula Formula* holds [4]:

$$P(X > 0, Y > 0) = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}$$

Explanation:

- The joint probability $P(X > 0, Y > 0)$ corresponds to the *orthant probability* in the positive quadrant.

- The term $\arcsin(\rho)$ arises from integrating the bivariate normal density over this quadrant.

Step 7. *Substitute Back to Find $\tilde{\rho}$.*

$$\tilde{\rho} = 4 \left(\frac{1}{4} + \frac{\arcsin(\rho)}{2\pi} - 0.25 \right) = \frac{2}{\pi} \arcsin(\rho)$$

B.2 LEMMA 2

$$|\tilde{\rho}| \leq |\rho|$$

Interpretation: Using the result of Lemma 1, the ratio $\frac{\tilde{\rho}}{\rho} = \frac{2}{\pi\rho} \arcsin(\rho)$ is always less than one for ρ in the interval $(-1, 1)$ except at $\rho = \pm 1$ where the ratio is equal to one.

Step 1

Properties of the Arcsine ($\sin^{-1}$) Function:

- The **arcsine function** $\arcsin(x)$ is defined for $x \in [-1, 1]$.
- The range of $\arcsin(x)$ is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.
- The function $\arcsin(x)$ is **increasing** on $[-1, 1]$.

Step 2

We can consider ρ in one of three intervals:

1. $\rho = 0$
2. $0 < \rho < 1$
3. $-1 < \rho < 0$

Step 3 (Case 1: $\rho = 0$)

- $\arcsin(0) = 0$
- The ratio becomes indeterminate ($\frac{0}{0}$), but we can consider the limit as $\rho \rightarrow 0$.
- As $\rho \rightarrow 0$, $\arcsin(\rho) \approx \rho$ (using the first-order Taylor expansion).
- Therefore, $\frac{\tilde{\rho}}{\rho} \approx \frac{2}{\pi\rho} \cdot \rho = \frac{2}{\pi} \approx 0.6366$.

Step 4 (Case 2: $0 < \rho < 1$)

Both ρ and $\arcsin(\rho)$ are positive.

Define the function:

$$f(\rho) = \frac{2}{\pi\rho} \arcsin(\rho)$$

We need to show that $f(\rho) < 1$ for $0 < \rho < 1$.

For $0 < \rho < 1$: $\arcsin(\rho) < \frac{\pi}{2}\rho$

This inequality holds because of the well known result that the function $\arcsin(\rho)$ is **strictly convex** on $(0, 1)$, and its graph lies below the straight line $y = \frac{\pi}{2}\rho$ [20].

Therefore: $f(\rho) < 1$ for all $0 < \rho < 1$.

Step 5 (Case 3: $-1 < \rho < 0$)

- When ρ is negative $\arcsin(\rho)$ is negative, also implying that $\tilde{\rho}$ is negative.
- The ratio $\frac{\tilde{\rho}}{\rho}$ is therefore positive because a negative divided by negative is positive.
- Since $\arcsin(\rho)$ is negative and increasing from $-\frac{\pi}{2}$ to 0 as ρ increases from -1 to 0, we can use logic similar to Case 2.
- Compute the Limit as $\rho \rightarrow -1^+$.
- $\arcsin(-1) = -\frac{\pi}{2}$
- $f(-1) = \frac{2}{\pi \times (-1)} \times \left(-\frac{\pi}{2}\right) = 1$

- For $-1 < \rho < 0$: $\arcsin(\rho) > \frac{\pi}{2}\rho$

(using the same logic as in case 2, except that the signs are opposite)

$f(\rho) < 1$ for all $-1 < \rho < 0$.

Step 6

- The ratio $\frac{\tilde{\rho}}{\rho} = \frac{2}{\pi\rho} \arcsin(\rho)$ is **less than one** for all ρ in $(-1, 1)$.
- The ratio equals **one only when** $\rho = 1$ or $\rho = -1$.
- Therefore, **the ratio is always less than or equal to one**.
- Furthermore, since $\tilde{\rho}$ will always have the same sign as ρ , the magnitude of $\tilde{\rho}$ will always be less than the magnitude of ρ

$$|\tilde{\rho}| \leq |\rho|$$

Implications

- Under this result, binarization reduces or at worst preserves correlation - never inflates it beyond its original absolute value. That is, for any pair of binarized features, the magnitude of correlation coefficient $\tilde{\rho}$ is strictly less than the original ρ , except at ± 1 where they are equal.
- This result effectively keeps all off-diagonal correlations from saturating to exactly ± 1 when no column is an exact multiple of another.
- If all off-diagonal entries of a correlation matrix are strictly bounded away from ± 1 , then the matrix's smallest eigenvalue $\lambda_{\min}(R)$ is always positive [9].
- Therefore, R^{-1} is well-defined and has a finite spectral norm bound given by:

$$\|R^{-1}\|_2 \leq \frac{1}{\lambda_{\min}(R)}.$$

- Hence, the boundedness of R^{-1} is implied when $|\tilde{\rho}| \leq |\rho|$ and $\tilde{\rho} \neq \pm 1$ and the entire correlation matrix must be invertible, with binarization universally enhancing its conditioning for more stable inversion in practice.
- Thus, a matrix-level corollary can be extended from the pairwise-level bound shown above. Both statements reinforce the same theme: binarization controls correlations and preserves invertibility in a way beneficial to sparse regression methods like LASSO.

B.3 LEMMA 3

$$|\text{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j})| \leq |\text{Cov}(x_{S^c i}, x_{S j})|$$

Interpretation: Binarization reduces the covariance between features for any irrelevant feature $x_{S^c i} \in X_{S^c}$ and relevant feature $x_{S j} \in X_S$.

Review of Assumptions:

1. **Joint Normality:** The continuous features $x_{S^c i}$ and $x_{S j}$ are jointly normally distributed with zero means and unit variances:

$$x_{S^c i}, x_{S j} \sim \mathcal{N}(0, 1)$$

and correlation $\rho_{ij} = \text{Corr}(x_{S^c i}, x_{S j})$.

2. **Binartization at Zero Threshold** - The binarized variables are defined as:

$$\tilde{X}_k = \begin{cases} 1 & \text{if } X_k > 0 \\ 0 & \text{if } X_k \leq 0 \end{cases}, \quad k = i, j$$

Step 1

The covariance between $x_{S^c i}$ and $x_{S j}$ is:

$$\text{Cov}(x_{S^c i}, x_{S j}) = \rho_{ij} \cdot \sigma_{x_{S^c i}} \sigma_{x_{S j}} = \rho_{ij} \cdot (1)(1) = \rho_{ij}$$

Step 2

The covariance between $\tilde{x}_{S^c i}$ and $\tilde{x}_{S j}$ is:

$$\text{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j}) = \mathbb{E}[\tilde{x}_{S^c i} \tilde{x}_{S j}] - \mathbb{E}[\tilde{x}_{S^c i}] \mathbb{E}[\tilde{x}_{S j}]$$

Since $x_{S^c i}$ and $x_{S j}$ are symmetric about zero:

$$\mathbb{E}[\tilde{x}_{S^c i}] = P(x_{S^c i} > 0) = \frac{1}{2}, \quad \mathbb{E}[\tilde{x}_{S j}] = \frac{1}{2}$$

Compute the joint probability:

$$\mathbb{E}[\tilde{x}_{S^c i} \tilde{x}_{S j}] = P(x_{S^c i} > 0, x_{S j} > 0)$$

For jointly standard normal variables, this joint probability is given by [4]:

$$P(x_{S^c i} > 0, x_{S j} > 0) = \frac{1}{4} + \frac{\arcsin(\rho_{ij})}{2\pi}$$

Therefore, the covariance is:

$$\text{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j}) = \left(\frac{1}{4} + \frac{\arcsin(\rho_{ij})}{2\pi} \right) - \left(\frac{1}{2} \cdot \frac{1}{2} \right) = \frac{\arcsin(\rho_{ij})}{2\pi}$$

Step 2

Comparing Covariances Before and After Binarization

We need to show that:

$$\left| \frac{\arcsin(\rho_{ij})}{2\pi} \right| \leq |\rho_{ij}|$$

Case 1: When $\rho_{ij} \geq 0$

The function $f(\rho) = \frac{\arcsin(\rho)}{2\pi}$ is increasing in $[0, 1]$. Since $\arcsin(\rho) \leq \frac{\pi}{2}$, we have:

$$0 \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq \frac{1}{4}$$

Also, $\rho_{ij} \leq 1$, so:

$$0 \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq \rho_{ij}$$

because $\arcsin(\rho_{ij}) \leq \frac{\pi}{2}\rho_{ij}$ in $[0, 1]$.

Step 3

Case 2: When $\rho_{ij} \leq 0$

The function $f(\rho) = \frac{\arcsin(\rho)}{2\pi}$ is increasing in $[-1, 0]$. Since $\arcsin(\rho_{ij}) \geq -\frac{\pi}{2}$, we have:

$$-\frac{1}{4} \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq 0$$

Also, $\rho_{ij} \geq -1$, so:

$$\rho_{ij} \leq \frac{\arcsin(\rho_{ij})}{2\pi} \leq 0$$

because $\arcsin(\rho_{ij}) \geq \frac{\pi}{2}\rho_{ij}$ in $[-1, 0]$.

Therefore, in both cases:

$$|\text{Cov}(\tilde{x}_{S^c i}, \tilde{x}_{S j})| = \left| \frac{\arcsin(\rho_{ij})}{2\pi} \right| \leq |\rho_{ij}| = |\text{Cov}(x_{S^c i}, x_{S j})|$$

Conclusion

The magnitude of the covariance between $\tilde{x}_{S^c i}$ and $\tilde{x}_{S j}$ is less than or equal to the magnitude of the covariance between $x_{S^c i}$ and $x_{S j}$. Therefore, binarization reduces the covariances between irrelevant and relevant features.

B.4 LEMMA 4

When $\tilde{\rho}$ and $\rho > 0$,

$$\|\tilde{G}^{-1}\|_{\infty} \leq \|G^{-1}\|_{\infty}$$

Interpretation: The Infinity norm of the binarized matrix is less than or equal to the Infinity norm of the original continuous matrix.

Definitions:

- $n > 0$: Positive scalar (e.g., sample size).
- $s \geq 2$: Dimension of the square matrices ($s \times s$).
- $\mathbf{I}_s$: Identity matrix of size s .
- $\mathbf{J}_s$: All-ones matrix of size s .
- $0 < \tilde{\rho} < \rho < 1$: Constants with $\tilde{\rho}$ always less than ρ .

Step 1

Original Gram Matrix:

$$G = n((1 - \rho)\mathbf{I}_s + \rho\mathbf{J}_s)$$

Binarized Gram Matrix:

$$\tilde{G} = n((1 - \tilde{\rho})\mathbf{I}_s + \tilde{\rho}\mathbf{J}_s)$$

Inverse of G and $\tilde{G}$:

First, rewrite G as

$$G = n(1 - \rho) \left(\mathbf{I}_s + \frac{\rho}{1 - \rho} \mathbf{J}_s \right)$$

Let $\alpha = \frac{\rho}{1 - \rho}$.

Compute $(\mathbf{I}_s + \alpha\mathbf{J}_s)^{-1}$:

- Observation: $\mathbf{J}_s$ is rank-1.
- Sherman-Morrison-Woodbury formula for the inverse:

$$(\mathbf{A} + \mathbf{UCV}^T)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^T\mathbf{A}^{-1}\mathbf{U})^{-1}\mathbf{V}^T\mathbf{A}^{-1}$$

- $\mathbf{A} = \mathbf{I}_s \Rightarrow \mathbf{A}^{-1} = \mathbf{I}_s$
- $\mathbf{U} = \mathbf{V} = \mathbf{1}_s$
- $\mathbf{C} = \alpha$
- $\mathbf{C}^{-1} = \frac{1}{\alpha}$ (the inverse of a scalar being its reciprocal)
- $\mathbf{V}^T\mathbf{A}^{-1}\mathbf{U} = \mathbf{1}_s^T\mathbf{1}_s = s$

For $\mathbf{I}_s + \alpha\mathbf{J}_s$:

$$(\mathbf{I}_s + \alpha\mathbf{J}_s)^{-1} = \mathbf{I}_s - \frac{\alpha}{1 + \alpha s} \mathbf{J}_s$$

Therefore,

$$G^{-1} = \frac{1}{n(1-\rho)} \left(\mathbf{I}_s - \frac{\rho}{1+\rho(s-1)} \mathbf{J}_s \right)$$

Similarly, for $\tilde{G}$:

$$\tilde{G}^{-1} = \frac{1}{n(1-\tilde{\rho})} \left(\mathbf{I}_s - \frac{\tilde{\rho}}{1+\tilde{\rho}(s-1)} \mathbf{J}_s \right)$$

Step 2

Definition of Infinity Norm for Matrices:

$$\|A\|_{\infty} = \max_{1 \leq i \leq s} \sum_{j=1}^s |a_{ij}|$$

Compute Entries of G^{-1} :

- Diagonal Entries: $(G^{-1})_{ii} = \frac{1}{n(1-\rho)} (1 - \gamma)$
- Off-Diagonal Entries: $(G^{-1})_{ij} = -\frac{\gamma}{n(1-\rho)}$ for $i \neq j$

Where: $\gamma = \frac{\rho}{1+\rho(s-1)}$

Compute the Row Sum for G^{-1} :

Row Sum:

$$\text{RowSum}_{G^{-1}} = \frac{1}{n(1-\rho)} [(1 - \gamma) + (s-1)\gamma]$$

Simplify the Row Sum:

$$\text{RowSum}_{G^{-1}} = \frac{1}{n(1-\rho)} [1 - \gamma + \gamma(s-1)] = \frac{1}{n(1-\rho)} [1 + \gamma(s-2)]$$

Where: $\tilde{\gamma} = \frac{\tilde{\rho}}{1+\tilde{\rho}(s-1)}$

Similarly:

$$\text{RowSum}_{\tilde{G}^{-1}} = \frac{1}{n(1-\tilde{\rho})} [1 + \tilde{\gamma}(s-2)]$$

Step 3

Establishing the Relationship between γ and $\tilde{\gamma}$

Define function $f(x) = \frac{x}{1+x(s-1)}$

Due to the monotonicity of $f(x)$, when $x_1 < x_2$, $f(x_1) < f(x_2)$

It follows that:

- $f(x)$ is *increasing* wrt x on $(\frac{-1}{s-1}, 1)$.
- $|\tilde{\rho}| \leq |\rho|$ (Lemma 2), and both are bounded by the interval $(-1, 1)$
- Since $\tilde{\rho} \geq \rho$ when $\tilde{\rho}$ and ρ are less than zero (meaning that ρ is *more* negative), the monotonicity of $f(x)$ implies that $|f(\tilde{\rho})| \leq |f(\rho)|$
- Therefore, we have $|\tilde{\gamma}| = |f(\tilde{\rho})| \leq |f(\rho)| = |\gamma|$, with a discontinuity at $x = \frac{-1}{s-1}$.
- Furthermore, $\tilde{\rho}, \rho$, and hence $\gamma, \tilde{\gamma}$ will all be of the same sign in the interval $[\frac{-1}{s-1}, 1)$ for $\tilde{\rho}$ and ρ , and γ will be positive otherwise.

Step 4

Comparing Row Sums:

Since $|\gamma| \geq |\tilde{\gamma}|$ and $s-2 \geq 0$ (because $s \geq 2$),

$$\gamma(s-2) \geq \tilde{\gamma}(s-2) \text{ when } \gamma > 0$$

Therefore,

$$1 + \gamma(s-2) \geq 1 + \tilde{\gamma}(s-2) \text{ when } \gamma > 0$$

Comparing Denominators:

Since $|\tilde{\rho}| \leq |\rho|$,

$$\bullet \quad 1 - \tilde{\rho} \geq 1 - \rho \Rightarrow \frac{1}{n(1-\tilde{\rho})} \leq \frac{1}{n(1-\rho)} \text{ when } \tilde{\rho}, \rho > 0$$

$$\bullet \quad 1 - \tilde{\rho} \leq 1 - \rho \Rightarrow \frac{1}{n(1-\tilde{\rho})} \geq \frac{1}{n(1-\rho)} \text{ when } \tilde{\rho}, \rho < 0$$

Step 5

Putting It All Together

Infinity Norms when $\tilde{\rho}, \rho$ and $\gamma > 0$:

$$\|G^{-1}\|_{\infty} = \frac{1}{n(1-\rho)} [1 + \gamma(s-2)]$$

$$\|\tilde{G}^{-1}\|_{\infty} = \frac{1}{n(1-\tilde{\rho})} [1 + \tilde{\gamma}(s-2)]$$

Conclusion

Since both terms examined in the expression for G^{-1} are greater than the corresponding terms in the expression for $\tilde{G}^{-1}$ when $\tilde{\rho}, \rho$ and γ are greater than zero,

$$\|G^{-1}\|_{\infty} \geq \|\tilde{G}^{-1}\|_{\infty}$$

Furthermore, when $s = 2$ and $\tilde{\rho}, \rho$ and γ are less than zero it can be shown that the reverse is guaranteed to be true. When $s > 2$ the answer is dependent upon where $\tilde{\rho}$ or ρ fall with respect to the interval $[\frac{-1}{s-1}, 1)$. When ρ is below the lower bound but $\tilde{\rho}$ is above, the relative norm values will be ambiguous. By constraining ρ and $\tilde{\rho}$ to be greater than zero, the inequality is unambiguous.

As a result, in the positive-correlation PoC regime the binarization process *enhances positive correlation relationships* and, when $\tilde{\rho}$ and ρ are on the same side of the discontinuity at $\frac{-1}{s-1}$, it

will also deemphasize negative correlation relationships (so, this will happen the vast majority of the time). Note that this seeming disadvantage can be mitigated by adding the complement to each binarized column into the feature matrix.

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2. 2025 IEEE International Conference on Big Data, "Efficient Longitudinal Feature Selection via Binarized Transformation: Theory and Case Studies", 08-11 December 2025
3. 2025 International Conference on Big Data Analytics & Applications (BDAA 2025), "Any-time Rule Compression and Rectified Logistic Modeling for Longitudinal Signals", 25-27 November 2025